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COS700 Research Report

A Blockchain-Enabled Framework for Misinformation Monitoring

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Abstract

Election misinformation in South Africa spreads faster than any single organisation can verify it, often in languages automated tools do not support, and corrections are frequently dismissed as biased. Using design-science research, this study built and evaluated a framework combining multilingual AI detection with a permissioned blockchain shared by fact-checking organisations. Because no labelled election corpus existed in Sepedi or isiZulu, statements were generated in English with a large language model and machine-translated, yielding 13,997 statements balanced across English, Sepedi and isiZulu. AfroXLMR-large-76L outperformed all other models tested, reaching a mean F1-score of 0.899 with full fine-tuning and 0.882 with LoRA, with Sepedi and isiZulu trailing English by about 0.04. The LoRA model, which overfitted less, flags claims for human fact-checkers, whose votes are recorded on a Hyperledger Fabric consortium ledger with roles enforced in chaincode through X.509 certificate attributes. Claim text and review evidence were held off-chain, with content commitments and review metadata recorded on-chain. Across consortium sizes of 3, 5, 10, 15 and 20 organizations, all 750 recorded feed-workflow observations completed and passed integrity verification. Mean ingestion-to-committed-claim latency remained approximately 2.39–2.44 seconds, while mean ingestion-to-final-verdict latency increased from 6.69 to 24.09 seconds as the required scripted review majority increased. Direct Caliper tests covered five consortium sizes and five per-round transaction counts from 100 to 2,000, with no reported write or read failures. At 2,000 requests per round, mean write and full-scan read throughput across consortium sizes were 136.52 and 27.02 transactions per second, respectively. These results describe synthetic inputs, scripted reviews and a single-host deployment.

Keywords:

election misinformation, multilingual misinformation detection, low-resource languages, AfroXLMR, LoRA, permissioned blockchain, Hyperledger Fabric, fact-checking, POPIA, South Africa

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1 Introduction

In the months before South Africa's 2024 general election, AI-generated deepfakes in which Joe Biden, Donald Trump and the rapper Eminem appeared to intervene in the campaign spread through politically active WhatsApp groups [1]. Gondwe [1] calls the 2024 elections "a watershed moment for misinformation research on the continent". When group members posted fact-checks, the corrections were "often met with suspicion, dismissed as propaganda, or reframed as a foreign attempt to meddle in domestic politics".

Those observations frame this study. False content now travels faster than any single organisation can verify it; the corrections that do appear are not trusted by the people who most need them; and the automated tools that could help with scale were not built for South Africa's languages. This report presents and evaluates a framework that addresses all three together: a multilingual AI model that flags likely misinformation in English, Sepedi and isiZulu, and a permissioned blockchain on which a consortium of fact-checking organisations records, votes on and publishes its verdicts in a form that no single member can alter. Following Wardle and Derakhshan, as summarised by Monnier and Ségur [2], disinformation is false content created to deceive and misinformation is false content shared without that intent; because a monitoring system rarely knows intent, this report uses misinformation as the umbrella term.

The institutional response to the 2024 election was substantial. The Electoral Commission of South Africa (IEC), Media Monitoring Africa, the major platforms, newsrooms and a civil-society fact-checking coalition all took part [3], [4], although the media had to cover the election "with limited resources" [5]. The result is an ecosystem of many capable actors, each keeping its own record of what it has checked and concluded. What it lacks is a shared, verifiable record that connects those conclusions. This study argues that this gap limits the ecosystem as much as any shortage of detection technology, and it treats the forthcoming local government elections as the next test.

Fact-checking itself works: in simultaneous experiments in Argentina, Nigeria, South Africa and the United Kingdom, fact-checks reduced belief in misinformation, and most of the effects were still visible more than two weeks later [6]. The weakness lies in the system around it. It is reactive, with the same "zombie claims" debunked every two years [7]. It is fragmented, with verdicts published on different rating scales, so that establishing whether two organisations agree has taken a dedicated research project rather than a simple query [8],

[9]. And it struggles to earn trust, as fact-checks are dismissed as propaganda or seen as partisan [1], [7], [10].

Automation is unavoidable at election scale, but in South Africa it runs into language. Social media platforms "do not adequately moderate non-English language content" [11], and in a review of misinformation detection research, 67% of studies used English datasets and fewer than 10% focused on low-resource languages [12]. South African studies have begun to close the gap in Tshivenda, Sesotho, Sepedi and isiZulu [13–16], but each machine-translated an English dataset, and none addresses elections. Nor can a classifier be the final word, since an unexplained machine verdict gives a sceptical audience one more reason to distrust. The model in this study therefore acts as a scout rather than a judge: it escalates likely misinformation to human fact-checkers, and its label and confidence score are recorded alongside their verdicts, not in place of them.

Because the core problem is that independent organisations cannot easily share, compare and trust each other's verdicts, the remedy must be organisational as well as technical. A permissioned blockchain gives all member parties a unified view of the data, while no single party can change it without the consensus of the others [17]. A ledger is not a source of truth, however, so in this framework human fact-checkers, not the model, stand between a claim and its final status on the ledger. The ledger must also respect the Protection of Personal Information Act (POPIA), recording decisions about content without becoming a permanent register of the people who wrote or shared it.

These topics of research seldom meet. Detection studies stop at the classifier, fact-checking research proposes organisational rather than infrastructural remedies, and blockchain designs for news verification are not built for low-resource languages or for verdicts shared among many organisations. As far as this study could establish, no existing work combines detection in South African languages, verdicts reached jointly by multiple independent organisations and a tamper-evident shared ledger, or measures how such a system performs as the number of member organisations grows.

The report is organised as follows. Section 2 states the problem and the research questions. Section 3 describes the methodology: the synthetic corpus, the detection model, and the design and testing of the blockchain layer. Section 4 provides the background, and Section 5 reviews related work and sets out the research gap. Section 6 discusses the results, and Section 7 concludes.

2 Problem Statement

2.1 Problem statement

Misinformation around elections poses a greater threat than ever to South Africa's local government elections in 2026. The scale of this challenge was evident during the 2024 national elections, where AI-generated deepfakes went viral, racking up hundreds of thousands of views on social media and WhatsApp groups. Although major South African news agencies such as News24, Daily Maverick and Africa Check have institutionalised fact-checking, the current landscape is fragmented and siloed. International literature on fact-checking organisations (PolitiFact, Snopes, FactCheck.org) indicates structural systemic weaknesses: inconsistent claim selection, varied rating systems, limited crossvalidation capacity and, crucially, a weakening of public trust, where corrections are often rejected as propaganda rather than credible information.

Although Transformer Models and Large Language Models have been studied to enable the automation of misinformation detection at scale, most of the work done has been on the English language, their effectiveness is still limited in low-resource, underrepresented South African languages due to insufficient annotated datasets. At the same time, misinformation monitoring remains fragmented across fact-checking organisations, media institutions and electoral stakeholders. This fragmentation limits the ability to securely record, trace, verify and audit misinformation reports across organisations.

Blockchain technology offers the opportunity to create a decentralised, transparent and immutable record of fact-checking decisions, accessed by independent fact checkers, election officials and civil society organisations to collaboratively verify claims while keeping an auditable ledger. However, there is currently no operationalisation of this strategy, especially in combination with an AI-enabled misinformation flagging system. The core research problem therefore lies in the absence of an integrated AI–blockchain approach to support multilingual misinformation detection, fact-checking transparency, accountability and trust in electoral contexts.

2.2 Research question

How can an integrated AI and blockchain-based framework be designed and evaluated to enable tamper-resistant, transparent, multilingual monitoring and verification of election misinformation within South Africa's multistakeholder electoral context.

2.3 Sub Research questions

1. **Artificial Intelligence enabled multilingual misinformation detection.**

   Which artificial intelligence enabled approaches are most suitable for detecting misinformation across multiple South African languages, and how can this be adapted to address low-resource data?

2. **Blockchain architecture to enable transparent, tamper-resistant, and auditable fact-checking.**

   Which blockchain architecture best balances data integrity, privacy, accessibility and scalability for a consortium-based fact-checking network in South Africa?

3. **Artificial Intelligence enabled misinformation detection and blockchain integration.**

   How can artificial intelligence enabled misinformation detection be integrated with a blockchain ledger to support an end-to-end, tamper-resistant and auditable monitoring workflow?

4. **Privacy and legal compliance.**

   Which privacy-preserving and cryptographic techniques can satisfy South Africa's Protection of Personal Information Act (POPIA) while maintaining the transparency and auditability of fact-checking decisions?

5. **Evaluation & trust.**

   To what extent can the integrated AI blockchain framework improve transparency, traceability and trust in multi-stakeholder misinformation monitoring, and how does it compare with traditional centralised approaches in terms of data quality, auditability and institutional accountability?

2.4 Research Objectives

The following objectives were pursued, each addressing one or more of the sub-research questions above. Section 6 reports the extent to which each was met.

Objective 1. To examine the challenges of detecting misinformation within multilingual, low-resource South African language contexts (SRQ1).

Objective 2. To investigate existing AI/NLP approaches for misinformation detection and assess their suitability for a multilingual South African election environment (SRQ1).

Objective 3. To design a blockchain architecture that balances data integrity, privacy, accessibility and scalability for a consortium based fact checking network (SRQ2).

Objective 4. To integrate the AI-based misinformation-detection component with the blockchain ledger to form an end-to-end, tamper-resistant and auditable monitoring workflow (SRQ3).

Objective 5. To implement a privacy-preserving on-chain/off-chain data model aligned with South Africa's Protection of Personal Information Act (SRQ4).

Objective 6. To implement a working scalable prototype demonstrating the technical feasibility of the integrated framework.

Objective 7. To evaluate the prototype against measurable criteria: AI detection quality (accuracy, precision, recall, F1), blockchain system performance (throughput, latency, scalability) and POPIA alignment (SRQ5).

Objective 8. To analyse the limitations, trade-offs and governance considerations of integrating AI and blockchain within a multilingual election misinformation monitoring system.

2.5 Scope and Limitations

This study does not address real-time, production-scale deployments against live social media platforms. The detection component is trained and evaluated on a synthetic, machine-translated corpus rather than live-scraped South African social media content. It does not address fully distributed, multi-institutional blockchain deployments across physically separated organisational servers. The ledger was benchmarked with up to twenty simulated organisational identities on a single host; therefore, network conditions between

physically separated institutional servers remain outside the scope of this study and are considered as future work.

The AI detection component covers English, Sepedi and isiZulu. A REST-based integration layer was implemented as the primary access point for stakeholder organisations, but a stakeholder-facing web dashboard or user interface was not implemented. A general-purpose blockchain explorer provides a read-only operational view rather than an end-user application interface.

3 Methodology

This study followed a design-science research (DSR) approach [18], [19] combined with a mixed-methods evaluation strategy. DSR suits problems in which the research objective is to construct and evaluate an innovative technological artefact that solves an explicit real-world problem; here, the lack of a verifiable, tamper-resistant and privacy-compliant audit trail for election misinformation in South Africa. Quantitative methods measure detection quality and blockchain performance (latency, throughput and scalability), addressing objectives 6 and 7, while qualitative analysis assesses governance structure, auditability and alignment with data protection legislation [20], addressing objective 5.

The subsections below follow the DSR cycle whereby Sections 3.1–3.2 cover artefact construction for the detection component, Sections 3.3–3.6 cover construction and justification of the blockchain component (SRQ2, SRQ3) and Section 3.7 covers the evaluation methods used against both (SRQ5).

3.1 Data Acquisition

3.1.1 Data Acquisition: Synthetic Data Generation

No labelled corpus of South African election misinformation exists in Sepedi or isiZulu. The closest South African studies built their training data by machine-translating an English dataset [13–16], but translating an existing English dataset would not have produced content about South African elections. This study therefore generated an election-specific corpus with large language models (LLMs) and translated it into Sepedi and isiZulu, an approach increasingly used to support low-resource classification where labelled data is scarce, expensive or sensitive [21]. Because every statement is synthetic, no personal information of real social media users was collected.

The corpus was produced in three stages: prompt construction, generation, and translation (Table 1). Each stage appended its output to a file instead of overwriting it, wrote rows it could not process to a separate failure file, and saved an audit record of the model, time and prompt range of each run, so every row in the final corpus can be traced to the prompt and model that produced it.

Table 1: Stages of the synthetic data pipeline.

Stage	Tool	Output
1. Prompt construction	Python script, fixed random seed	5,000 prompts, each requesting 10 labelled English samples
2. Generation	Gemini 3.6 Flash	Statements with a label, a claimed source and an explanation
3. Translation	Gemini 3.6 Flash	Each statement in one of English, Sepedi or isiZulu

3.1.1.1 Prompt construction

Prompts were built by a script rather than written by hand, so that the corpus would cover the space of election misinformation systematically and could be regenerated exactly. The script produced 5,000 prompts from a single recorded random seed (797973867). Each prompt combines one value from each of the axes in Table 2. It also sets four of twelve style features, each to one of four levels. The features are sarcasm, formality, emotional intensity, certainty, sentence complexity, vocabulary level, perspective, rhetorical devices, urgency, South African English colouring, punctuation style and attitude to authority.

Table 2: Prompt axes. One value was drawn from each axis for every prompt.

Axis	Values	Examples
Topic	25	voters' roll and registration; special votes; ballot-box chain of custody; coalition talks; load-shedding at voting stations; fake IEC statements
Format	10	WhatsApp forward; transcribed voice note; tweet; Facebook community post; news headline and article; SMS blast; community-radio comment
Setting	8	metro; township; small rural town; village under traditional authority; informal settlement; mining-belt municipality
Tone	8	angry and accusatory; concerned citizen; sarcastic; conspiratorial; neutral news report
Election period	5	2021 and 2024 elections (retrospective); 2026 local government campaign; voting day (4 November 2026); counting and results
Misinformation mechanism	8	fabricated event; real process with false details; true fact with a false explanation; manipulated statistic; false attribution; recycled old incident
Accurate content	8	voter-education information; factual news report; accurate description of IEC procedure; opinion framed as opinion; correct debunk of a rumour

Each prompt asked for ten English samples, five labelled *misinformation* and five *non-misinformation*, all sharing the prompt's topic, format, setting, tone and style. Because

both classes are written under identical conditions, a classifier cannot separate them by topic or writing style alone; it has to learn the difference between false and accurate claims. The prompt instructed the model that every statement must make a different claim; that misinformation must be plausible and written in the voice of someone who believes it, and that non-misinformation must be accurate about how South African elections work.

3.1.1.2 Generation

- **Gemini 3.6 Flash** processed prompts 1–1,530 through the Gemini Batch API and returned 16,535 samples. Google Search grounding was enabled so that the model could base the *source* field on real reporting.

3.1.1.3 Translation

Rows were assigned to languages in rotation by position (isiZulu, Sepedi, English), so each statement appears in exactly one language and the three languages are roughly equal in size. English rows were kept as generated. The others were translated by Gemini 3.6 Flash, which was instructed to translate the full text, to preserve meaning, tone and register, to copy proper nouns, numbers, dates and hashtags unchanged, and to add or remove nothing. Rows that failed translation (978 in total) were excluded.

3.1.1.4 Quality control

Structure and traceability. Every model response was parsed as JSON, and rows that failed parsing were isolated rather than repaired. After translation, each row received an identifier built from hashes of its topic and statement and the identifier of the prompt that produced it, so that statements from the same prompt can be grouped.

Duplicates. No English source samples occur more than once.

Translation spot check. A random sample, stratified by language and label (seed 559243982), was back-translated into English with Gemini 3.6 Flash and compared with the original English statements. The check was completed for 50 samples: 25 Sepedi and 25 isiZulu. Human quality control was done on the 50 back translated samples, and results were satisfactory.

3.1.1.5 Final corpus

The final corpus contains 13,997 statements, balanced across languages and labels (Table 3).

Table 3: Final corpus by language and label.

Language	Misinformation	Non-misinformation	Total
English	2,360	2,310	4,670
Sepedi	2,360	2,307	4,667
isiZulu	2,341	2,319	4,660
Total	7,061	6,936	13,997

3.1.1.6 Limitations of the synthetic corpus

- **Realism.** Synthetic text may not match the vocabulary, errors and distribution of real posts, and LLM-generated data can contain factual inaccuracies and amplify bias [21]. The test set is also synthetic, so the results measure performance on generated statements; performance on real election posts has still to be established.
- **Labels.** Labels were assigned by the generating model and not verified by people. The identical isiZulu pair with opposite labels is one visible instance of label noise.
- **Translation.** Translation can change sentence structure and wording, although in the back-translation check of 50 samples Gemini 3.6 Flash preserved meaning and connotation.
- **Non-parallel languages.** Each statement exists in only one language, so per-language results compare different statements rather than the same statements in three languages.

3.2 Misinformation Detection: Model Selection and Evaluation

3.2.1 Misinformation Detection: Fine-Tuning and Model Selection

The detection component of the framework is a fine-tuned transformer classifier that labels each incoming statement as *misinformation* (the positive class) or *non-misinformation*. Its role is triage: it flags statements for human fact-checkers and does not issue verdicts itself.

The fine-tuning study had to settle two choices: which model to deploy, and whether to deploy a fully fine-tuned model or a lightweight adapter on a shared base model. It therefore compared a sequence of models of increasing capacity, all trained and tested on the same data split: three classical TF-IDF baselines, a South African decoder-only language model (MzansiLM-125M), and two African-language encoders (AfroXLMR-base-76L and AfroXLMR-large-76L). Each transformer was fine-tuned in two ways, updating all of its

parameters (full fine-tuning) and training only low-rank adapters (LoRA), and each combination was trained with three random seeds.

3.2.1.1 Data preparation

Text cleaning. Each statement in the synthetic corpus (Section 3.1.1) was cleaned in the following order:

1. remove emojis;
2. replace runs of exclamation marks with a full stop;
3. remove all punctuation except full stops;
4. collapse repeated whitespace;
5. remove opening sentences shorter than ten words;
6. convert to lower case.

Statements shorter than three characters after cleaning (100) and 20 degenerate statements were removed, leaving 13,877: 7,008 misinformation and 6,869 non-misinformation, with 4,661 in English, 4,642 in Sepedi and 4,574 in isiZulu.

Split. The statements were split 80/10/10 into training (11,099), validation (1,385) and test (1,393) sets, stratified by label and language (Table 4). Within each of the six label–language groups, statements kept their order in the corpus file, which is grouped by prompt, and were cut into consecutive training, validation and test segments; each segment was then shuffled. Statements from the same prompt therefore stay together: only 22 of the 1,408 prompts contribute statements to more than one split, which limits the leakage that near-identical statements written under the same prompt would otherwise cause. Which statements fall in which set is identical for every seed; the seed changes only their order. The validation set was used to choose the training epoch and to stop training early, and the test set only for the final evaluation.

Table 4: Training, validation and test sets.

Set	Total	Non-misinfo.	Misinfo.	English	Sepedi	isiZulu
Training	11,099	5,494	5,605	3,728	3,713	3,658
Validation	1,385	686	699	465	463	457
Test	1,393	689	704	468	466	459

3.2.1.2 Models

Classical baselines. Three classifiers were trained on TF-IDF features of word unigrams and bigrams, using terms that occur in at least two training statements and sublinear term-frequency scaling. The classifiers were logistic regression and a linear support vector machine (SVM), both with class weights balanced to the label distribution, and multinomial naive Bayes. For the SVM, which does not output probabilities, a softmax over its decision score was used so that ROC AUC could be computed. Traditional classifiers of this kind are the usual starting point in South African misinformation and hate-speech studies [15, 16], and they show how much the transformers add.

Transformers. Table 6 lists the three transformer models.

- **MzansiLM-125M** is a decoder-only model trained from scratch on a corpus covering all eleven official South African languages [22]. It is the only model in the study built specifically for South African languages. It also tests whether a small generative architecture can compete with encoders on classification. For classification, the hidden state of the last non-padding token was passed to the classification head. The end-of-sequence token served as the padding token, and padding was added on the right.
- **AfroXLMR-base-76L and AfroXLMR-large-76L** are versions of the multilingual encoder XLM-R adapted to African languages by continued masked-language-model training on African-language text [23]. The 76L versions cover 76 languages widely spoken in Africa, including Sepedi and isiZulu. On sentiment classification in South African languages, AfroXLMR outperformed other multilingual models and zero-shot GPT-4 [24]. Adapting it further to social media text improves results on classification tasks [25]. Fine-tuned BERT-style encoders generally outperform decoder-only large language models on classification [26], so the encoders were expected to lead.

Table 6: Transformer models.

Model	Architecture	Pre-training languages
MzansiLM-125M	Decoder-only, 125M parameters	All 11 official South African languages
AfroXLMR-base-76L	Encoder (XLM-R base)	76 languages widely spoken in Africa
AfroXLMR-large-76L	Encoder (XLM-R large)	76 languages widely spoken in Africa

3.2.1.3 Fine-tuning

Each transformer received a sequence-classification head with two outputs. Inputs longer than 256 tokens were truncated, and batches were padded to their longest sequence. Training used the AdamW optimiser with weight decay 0.01, a linear learning-rate schedule with warm-up, and 16-bit mixed precision. Training ran for up to ten epochs, and the model was evaluated on the validation set after each one. The checkpoint with the highest validation F1-score for the misinformation class was kept, and training stopped early when that score had not improved for two consecutive epochs. The loss was cross-entropy weighted by class,

$$w_c = \frac{N}{Kn_c} \tag{1}$$

where N is the number of training statements, $K = 2$ is the number of classes and n_c is the number of training statements in class c. Because the classes are nearly balanced, both weights are close to one. Table 7 gives the settings that differ between models. AfroXLMR-large used a smaller per-device batch with gradient accumulation and gradient checkpointing to fit in GPU memory, keeping an effective batch size of 16.

Table 7: Fine-tuning settings by model.

Model	Batch	Accumulation	Learning rate	Warm-up	Gradient checkpointing
MzansiLM-125M	16	1	2×10^{-5}	10%	No
AfroXLMR-base-76L	16	1	2×10^{-5}	10%	No
AfroXLMR-large-76L	8	2	1×10^{-5}	15%	Yes

Full fine-tuning and LoRA. Full fine-tuning updates every parameter of the model. Low-Rank Adaptation (LoRA) instead freezes the pre-trained weights and learns a low-rank update $\Delta W = BA$ for selected weight matrices, which cuts the number of trainable parameters sharply [27]. LoRA was applied to the query and value projections of every attention layer, with rank $r = 64$, scaling factor $\alpha = 128$ (a scale of $\alpha/r = 2$) and dropout 0.1, and without training bias terms. The classification head remained fully trainable. Under this configuration LoRA trains between 1.00% and 2.54% of each model's parameters (Table 8). Both methods used the settings in Table 7. Setting $\alpha = 2r$ follows Shuttleworth et al., who found that it produced fewer intruder dimensions and better generalisation than a smaller α [28].

Table 8: Trainable parameters under LoRA ($r = 64$, $\alpha = 128$). Totals include the adapters and the classification head.

Model	Trainable	Total	Trainable (%)
MzansiLM-125M	3,257,344	128,266,752	2.54
AfroXLMR-base-76L	2,951,426	296,356,612	1.00
AfroXLMR-large-76L	7,343,106	567,235,588	1.29

The comparison has a practical motivation. Parameter-efficient methods lower the cost of training and deploying models when data and compute are limited [29]. That matters for fact-checking organisations with small budgets, and it allows one base model to serve several tasks with small task-specific adapters. However, LoRA and full fine-tuning produce structurally different weight updates even when their accuracy is similar [28], so the study measures the difference rather than assuming the two are equivalent.

Seeds. Every combination of transformer and fine-tuning method was trained with three seeds (619, 436 and 777), giving 18 fine-tuning runs. The seed controls the initialisation of the classification head and the LoRA matrices, dropout, and the order in which training statements are presented. Multiple seeds were used because fine-tuned AfroXLM-R models vary considerably between runs in low-resource settings [30], so a single run can overstate or understate a difference between methods. The classical baselines are deterministic, and because the split is the same for every seed they give identical results in every run.

3.2.1.4 Evaluation

All models were evaluated once on the test set, after training had finished. A statement was predicted as misinformation when its predicted probability for that class was the higher of the two. The following measures were reported:

- accuracy;
- precision, recall and F1-score for the misinformation class;
- the area under the ROC curve (AUC), computed from the predicted probabilities.

F1-score is the primary evaluation measure and the measure used to select training checkpoints. Recall and precision both matter for a triage policy that escalates only predicted misinformation: a false negative would escape escalation, while a false positive would consume reviewer time. The feed benchmark used a different evaluation procedure, submitting both predicted labels to scripted review so that the full workflow could be measured. Its completion and label-overturn counts therefore do not measure the effectiveness of an escalation policy.

Results were also broken down by language (English, Sepedi and isiZulu), with Sepedi and isiZulu grouped together as African languages, to test whether performance holds for the lower-resource languages. For each model and method, the three seeds are summarised by the

mean and sample standard deviation of the test F1-score. The gap between methods is reported as $\Delta F_1 = \overline{F}_1^{full} - \overline{F}_1^{LoRA}$.

To compare how the two methods train, the analysis records the epoch chosen by early stopping in each run and the training and validation losses at that epoch. A validation loss well above the training loss indicates overfitting.

3.2.1.5 Selecting the deployed model

The model to deploy was chosen on the misinformation-class F1-score averaged over the three seeds, with the per-language results used to check that performance on Sepedi and isiZulu, on which most of the framework's value depends, did not fall far behind English. The choice between full fine-tuning and LoRA weighed F1 against cost: a LoRA adapter trains about 1–3% of the parameters and is a small file loaded on top of a shared base model, whereas a fully fine-tuned model has to be stored and served whole. The seed-to-seed spread and the training-loss analysis were also taken into account, because a method whose results vary widely from run to run is harder to rely on in deployment.

AfroXLMR-large-76L produced by seed 777 was retained as the deployed model. Its trained weights and tokenizer were saved at the end of fine-tuning and loaded into the detection component of the framework. For each statement it screens, the deployed model outputs a predicted label and its probability for that label. Both are recorded alongside the fact-checkers' votes (Section 3.5.1).

Implementation. The classical baselines were implemented with scikit-learn, and the transformers with PyTorch, Hugging Face Transformers (version 4) and the PEFT library. Training ran on a GPU in Google Colab.

3.3 System Architecture

The system is a decoupled, multi-service architecture with two layers, an Application and Ingestion Layer and a Permissioned Blockchain Persistence Layer (Figure 1), implementing objective 4 (end-to-end integration, SRQ3). Ingestion, machine-learning inference and blockchain submission are decoupled through an asynchronous Apache Kafka event-streaming pipeline [31], because election traffic surges unpredictably and direct synchronous submission from social media scrapers to a blockchain gateway would create immediate backpressure and request timeouts [32]. Kafka's publish–subscribe model

guarantees message durability, partition tolerance and rate smoothing between claim ingestion, classification and ledger commitment [33].

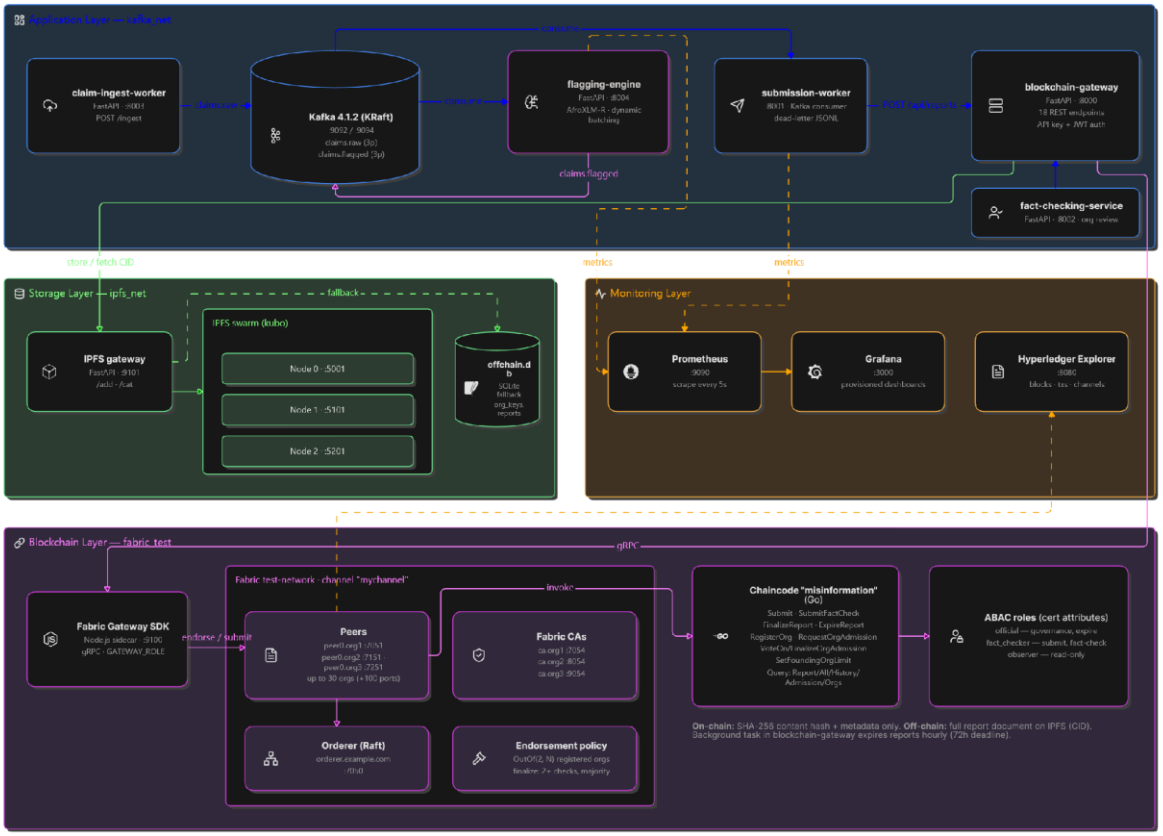


Figure 1: System Architecture Diagram

3.3.1 Blockchain Layer

A single automated setup procedure brings up, in order, the permissioned blockchain network, a peer-to-peer content-addressed storage swarm with a thin storage access service, a software development kit (SDK) sidecar that mediates all interaction with the ledger, and a REST gateway that is the sole point of contact for stakeholder organisations. The founding consortium size is configurable, and organisations can be admitted to a running network without a full reset: admission extends the channel configuration, re-commits the contract logic under an updated endorsement policy and re-keys the gateway services with the new organisation list.

3.3.2 Application Pipeline

An ingestion service publishes raw claim text to a message topic. A detection service consumes this topic, runs the transformer classifier (Section 3.2) with dynamic request

batching, in which a configurable maximum batch size or maximum queueing delay trades latency against throughput, and publishes labelled claims to a second topic. A submission service submits each labelled claim as a report to the blockchain gateway, and a fact-checking service gives each organisation a review interface over pending claims. Metrics from the pipeline services are collected and visualised through a standard monitoring stack.

3.4 HyperLedger Fabric Blockchain Framework

The persistence layer uses Hyperledger Fabric [34], a permissioned consortium blockchain framework. Two design decisions, addressing SRQ2 and Objective 3, are justified below: choosing a permissioned rather than a permissionless blockchain such as Bitcoin or Ethereum (Section 3.4.1), and choosing Hyperledger Fabric over the other permissioned frameworks (Section 3.4.2).

3.4.1 Permissioned Blockchain Justifications

No transaction fees. Permissionless blockchains such as Bitcoin [35] and Ethereum [36] impose variable, unpredictable transaction fees, called gas fees in proof-of-stake systems [37] such as Ethereum and miner fees in proof-of-work systems such as Bitcoin, which make continuous election monitoring financially unviable for non-profit fact-checkers and public institutions.

Institutional identity and non-repudiation. Election integrity requires strict institutional accountability. Fabric's permissioned Membership Service Provider (MSP) issues X.509 digital certificates to verified organisations, so every submission and vote is tied to a verified entity rather than an anonymous public key [34], [38].

Throughput. The system must keep pace with the velocity at which social media data arrives [39]. Permissionless consensus introduces probabilistic finality and high confirmation latency, whereas permissioned frameworks allow specialised ordering services with sub-second block confirmation, deterministic finality and no risk of chain forks [40], [41].

3.4.2 HyperLedger Fabric Justification

Hyperledger Fabric, R3 Corda, MultiChain and Quorum are consistently grouped together as the leading enterprise permissioned frameworks, and a systematic comparison across community activity, performance, scalability, privacy and adoption found that the choice between them is almost entirely use-case specific, with each option carrying its own trade-offs [42]. For this system the deciding requirement is that role checks are enforced in

chaincode, which the other three frameworks cannot provide.

Chaincode-level role enforcement. In a consortium the gateway is not a trusted boundary: each organisation could run its own gateway instance and modify it to skip role checks, or bypass it by submitting transactions directly to a Fabric peer or the SDK sidecar. Gateway-level enforcement would make the security of the whole network depend on every organisation behaving honestly, defeating the purpose of using a blockchain among mutually distrusting parties. In Fabric's execute–order–validate model, a transaction must be endorsed by a policy-defined set of peers before it can be committed. With role checks in chaincode, every endorsing peer independently evaluates the role attribute in the caller's X.509 certificate through Fabric's Client Identity (CID) chaincode library [34], [43]; if the caller lacks the required role, endorsement fails and the transaction is never ordered or committed, regardless of which client submitted it.

R3 Corda. Corda's privacy model is built on point-to-point communication: each message is addressed only to specific counterparts, there is no global broadcast of transactions, and only the parties involved build, validate and store a transaction [44]. An observer who is not a counterparty could not audit it, which rules out the network-wide, independent observer role that the system's auditability requires (Section 3.6).

Quorum. Quorum's enhanced, smart-contract-based permissioning lets a network administrator propose and approve new organisations and manage nodes, accounts and account-level access [45]. It is, however, a system-level administrative layer that governs who can join the network and transact, and it cannot evaluate role attributes from a participant's X.509 certificate inside application chaincode.

MultiChain. MultiChain grants global permissions to addresses (public-key or script hashes) through grant and revoke transactions [46]. MultiChain 2.0 added custom permissions that smart filters can enforce, but this remains a manual workaround rather than a native mechanism for embedding roles in participant identities and evaluating them in chaincode [46]. Its address-based, global model fits the system's three-role separation poorly, so MultiChain, like Corda and Quorum, was excluded.

3.5 Chaincode and On-Chain Data Model

To prevent ledger bloat, comply with South African data protection law and keep the system scalable, raw text is never stored on the blockchain. Raw claim text, full articles, AI inference

metadata, fact-checker reasoning and supporting evidence reside off-chain in an InterPlanetary File System (IPFS) Kubo swarm [47], and only structured metadata and SHA-256 report hashes are anchored on-chain, as in similar projects [38]. Storing large documents on a blockchain multiplies storage overhead across all full nodes and creates scalability bottlenecks [48], [49], and ledger immutability conflicts with POPIA's minimality and retention requirements (sections 10 and 14) [20], as discussed in Section 4.15; keeping personal data off the ledger satisfies POPIA at the architectural level. The design addresses objective 3 (SRQ2), objective 5 (SRQ4) and objective 6.

3.5.1 Report Identity and Off-Chain Storage

In the evaluated schema, the gateway derives a report identifier from the submitting organisation's MSP identifier and the ingestion message identifier using SHA-256. The claim document has a separate IPFS content identifier (CID) and SHA-256 content hash. The on-chain report records those commitments together with the AI-proposed label and confidence, model-version identifier, submitting organisation, submission time, 72-hour review deadline, status, eligible-reviewer set and required-vote threshold. Each review commits its outcome and the CID and SHA-256 hash of its supporting evidence. The verification endpoint checks retrieved content against the recorded commitments. Verification establishes consistency with the committed content; it does not establish whether a claim or a reviewer's decision is factually correct[50].

3.5.2 Consensus And Finalisation Paths

The ordering service uses Raft, a crash fault tolerant (CFT) consensus protocol in which a leader–follower model among ordering nodes establishes a total order of transactions before they are committed to the ledger [51]. Raft is Hyperledger Fabric's recommended ordering service because it is easier to configure and operate than the alternatives while providing deterministic finality and eliminating the risk of chain forks [52]. It was preferred over Byzantine fault tolerant (BFT) protocols for two reasons. All participating organisations are known and vetted, so the threat model is crash failures and transient network partitions rather than arbitrary malicious behaviour [53]; and in a direct comparison in Hyperledger Fabric, Raft achieved lower transaction latency and higher throughput than Practical Byzantine Fault Tolerance (PBFT) [54], [55]. Raft does not tolerate Byzantine faults [56], but the permissioned governance model and the MSP-issued X.509 identities, which tie every transaction to a verified entity, reduce the likelihood of Byzantine behaviour [34].

Quorum threshold. At submission, the report freezes its eligible-reviewer set as the registered organizations other than the submitting organization. For E eligible reviewers, the required threshold is floor(E/2) + 1 votes for the same label. This application-level review threshold is distinct from the two-organization endorsement policy and from ordering-service consensus.[51]

Finalization paths. A report automatically reaches FINAL when either label obtains the required majority. If every eligible reviewer votes without either label reaching that threshold, the report becomes DISPUTED and receives no final factual label. The evaluated workflow has no manual plurality-based report-finalization path. In the benchmark, scripted votes were submitted sequentially only until the required majority was reached.

3.6 Role Based Access Controls (RBAC)

Privileges are enforced through role-based access control (RBAC). Each identity receives a role attribute in its X.509 certificate when it enrols with the network's certificate authority, and the contract logic verifies this attribute through Hyperledger Fabric's Client Identity chaincode library before permitting any state-changing action [34] (Section 3.4.2). This addresses SRQ2 (privacy and accessibility trade-offs in the consortium's architecture). Three roles are defined:

- **Official**. May submit claims, submit a fact-check when its organisation is an eligible independent reviewer, expire eligible overdue reports, vote on and finalise organization-admission proposals, and configure the initial founding-organization limit before registration starts. Report verdicts are finalized automatically by the required reviewer majority.
- **Fact-checker.** May submit reports and fact-checks.
- **Observer.** Restricted to read-only queries of the ledger, whose blocks hold only SHA-256 content hashes and structured metadata, while the full claim and its evidence are held off-chain and addressed by content identifier. The role lets independent auditors examine what the registered organisations have classified as misinformation, adding an independent layer of transparency, the overall goal of the system.

3.7 Testing and Verification

Correctness checks are implemented as chaincode unit tests and REST-gateway integration tests. They cover report submission, independent-reviewer eligibility, automatic majority decisions for either label, disputed outcomes, expiry, duplicate and conflicting requests, evidence-integrity verification and role restrictions. These checks exercise the implemented contract and gateway behavior; the performance measurements below evaluate separate feed and direct-ledger workloads.

The primary benchmark exercised the complete Kafka → classifier → off-chain claim storage and Fabric submission → scripted reviewer voting → integrity-verification workflow. Consortium sizes of 3, 5, 10, 15 and 20 organizations contributed 150 workflow observations each, giving 750 observations overall. All runs used seed 100 and the same recorded source-data fingerprint, model checkpoint and execution-source profile. The records are processing observations rather than independent draws from real social-media traffic.

The benchmark runner used dataset-label mode where eligible organizations supplied sequential scripted votes for the synthetic dataset label until a strict majority was reached. All claims, including those proposed as non-misinformation, passed through the measured review workflow. Reviewer decisions were scripted rather than provided by human fact-checkers.

Three timing measures were recorded. Ingestion-to-committed-claim latency began at the HTTP ingestion request and ended when polling detected the committed claim. Ingestion-to-final-verdict latency began at the same request and ended when the authoritative FINAL report was retrieved after voting. The latter includes the scripted voting sequence but excludes the subsequent integrity-verification request, whose duration was measured separately. Service startup, backlog checks and invalid-message recovery probes were outside the sample-timing window. These definitions distinguish claim commitment, completed review and integrity verification.

The secondary benchmark used Hyperledger Caliper to measure direct ledger writes and full scan reads [57], [58]. This benchmark crossed 3, 5, 10, 15 and 20 organizations with per-round counts of 100, 250, 500, 1,000 and 2,000 transactions. Each configuration executed a write round followed by a read round, with two workers sharing the requested count. The fixed-load controller targeted 100 pending transactions; it did not impose fixed send rates of 25 TPS for writes or 50 TPS for reads. Send rate and throughput were observed separately. The direct workload used the Org1 client identity and excluded REST ingestion, Kafka, AI inference, IPFS access and reviewer voting.

The recorded benchmarks used Hyperledger Fabric 2.5.16, LevelDB world state, TLS-enabled connections, one Raft orderer, a two-organization endorsement policy, a 2-second block timeout and a maximum of ten messages per block. Both benchmark paths ran on the single host described in Table 9. Caliper latencies are reported in seconds and rates in transactions per second, the feed table reports mean durations in milliseconds.

Table 9: Benchmark device specifications.

Specification	Value
CPU	AMD Ryzen 7 5800HS (8 cores / 16 threads)
RAM	10GB (+ 10GB swapfile)
Platform	WSL2 on Windows (Microsoft hypervisor)
OS	Linux 6.18.33.2-microsoft-standard-WSL2
Flagging engine device	CPU (int8 quantization)

4 Background

4.1 Evolution of Election Misinformation in South Africa

South Africa's electoral framework already anticipates the problem of false political communication. Section 89(2) of the Electoral Act 73 of 1998 prohibits the publication of "any false information" with the intention of "disrupting or preventing an election", of "creating hostility or fear in order to influence the conduct or outcome of an election", or of otherwise influencing how an election is conducted or decided [4], [59]. Item 9 of the Electoral Code of Conduct similarly prohibits registered parties and candidates from publishing "false or defamatory allegations in connection with an election" [59]. These provisions were designed for an environment in which political messages came from identifiable speakers and travelled mainly through rallies, printed material and the mass media. Even in that environment, the boundary between a false statement of fact and legitimate political opinion was contested. In *Democratic Alliance v. African National Congress*, the Constitutional Court considered a campaign SMS sent to more than 1.5 million voters in Gauteng before the 2014 elections, and a key issue was whether the message was a statement of fact or an expression of opinion [59].

The 2024 national and provincial elections marked a significant change in this environment. The African National Congress (ANC) lost its national majority for the first time since 1994. The uMkhonto weSizwe Party (MKP), formed six months before the election, received about 15% of the national vote, and a government of national unity of ten parties followed [60]. Mare describes the elections as taking place amid the platformisation, digitisation and datafication of the media industry, declining trust in the media and the rise of social media influencers. Mare also identifies 2024 as the first South African election in which TikTok and AI-generated deepfakes "took the centre stage" [61]. Allen and le Roux identified an active and complex market for online influence in South Africa, in which some participants were motivated by ideology and others by money, and many influencers were paid to push key narratives [60].

Platform recommendation systems contributed to this shift from identifiable propaganda towards algorithmically amplified disinformation. Facebook's ranking emphasises content that generates comments and shares, which favours controversial material and can reinforce information silos into which fact-checked or corrective information struggles to penetrate [60]. TikTok's 'For You' page pushes content from accounts that users do not follow more

aggressively than other platforms, allowing new or manipulated information to reach large audiences rapidly regardless of its accuracy [60]. During the 2024 campaign, disinformation was planted well before polling day, preparing supporters to contest the result if it went against their party. Narratives portrayed the Electoral Commission of South Africa (IEC) and the electoral system as corrupt, technical glitches were framed as deliberate attempts to sway the result, and video or audio clips stripped of their context were shared across platforms as supposed proof of vote rigging [60]. On 25 May 2024, for example, TikTok and WhatsApp videos of IEC boxes at a KwaZulu-Natal venue circulated with claims that the boxes held improperly stored ballot papers; the IEC later clarified that the boxes contained bulk materials for setting up voting stations and no ballots [4], [60].

Encrypted messaging adds a further layer. WhatsApp accounts for the largest share of South Africa's social media users, but its encryption makes it difficult to research [60]. Senior editors worried that WhatsApp conversations and groups in vernacular languages may have done more to spread false information during the elections than X and Facebook [60]. During the 2024 elections, Gondwe analysed 22,384 messages posted by 6,283 users in 47 politically active WhatsApp groups. The disinformation relied most often on fear-based appeals (41%), identity-based rhetoric (32%) and content imitating credible journalism (27%), and it spread because it struck an emotional chord and was shared among trusted members of like-minded groups, not because it was accurate [1]. Together, these studies show that election misinformation in South Africa has moved from identifiable propaganda towards networked, algorithmically amplified and partly hidden disinformation, which the existing legal and institutional responses were not designed to monitor at scale.

4.2 Generative AI, Deepfakes and Synthetic Media: From the 2024 Elections to the 2026 Local Government Elections

Generative artificial intelligence has introduced synthetic media into this environment. Gondwe, drawing on Chesney and Citron, describes deepfakes as AI-generated synthetic media, from video and audio clips to images and text, that are increasingly used to manufacture inflammatory political material [1]. Small reports that AI amplified informational threats during the 2024 elections by artificially replicating the voices and videos of public figures. Because this form of informational manipulation was diffuse, remote and virtual, traditional static security arrangements were not fit for purpose [4].

Several cases from the 2024 elections illustrate the pattern. An AI-generated video in which Donald Trump appeared to endorse the MKP surfaced on TikTok more than a month before election day. Prominent MKP supporters amplified it, and it spread widely on X before Africa Check labelled it fake [60]. According to Africa Check, an AI-generated video in which the rapper Eminem appeared to endorse the Economic Freedom Fighters (EFF) attracted tens of thousands of views on Instagram, TikTok, X and YouTube [60]. Within WhatsApp groups, Gondwe documented a voice-cloned and lip-synced deepfake of Joe Biden warning of sanctions if the ANC won, the Trump and Eminem videos, and deepfake audio clips purportedly featuring Julius Malema [1]. Gondwe reports more than 158,000 views for the Trump video and 173,000 for the Eminem video, but describes these as upper-bound estimates of potential reach, based on the membership of the WhatsApp groups in which the videos appeared and on view counts from other platforms, rather than counts of unique viewers; audience overlap means some double-counting is likely [1]. These figures are therefore not directly comparable with the open-platform figure for the Eminem video [60]. These videos continued to circulate even after being flagged as false, and many users dismissed corrections as "mainstream manipulation" or foreign interference [1].

Assessments of the scale of AI use nevertheless differ. Allen and le Roux, whose analysis focused primarily on Facebook and X, concluded that the use of generative AI and deepfakes was limited but that more focused research was urgently needed [60]. Gondwe, by contrast, described WhatsApp groups as "saturated with highly convincing AI-generated content" [1]. This discrepancy is itself significant: synthetic media that circulates in encrypted, closed groups is largely invisible to the monitoring methods applied to open platforms, so its prevalence is difficult to establish. Audio is a particular blind spot. Only about 2% of the misinformation detection studies reviewed by Rananga et al. examined speech or audio, even though advances in audio editing make manipulated recordings increasingly difficult to distinguish from genuine ones [12]. Generative AI tools also raise concerns because their outputs can be convincing while lacking credible references or containing hallucinated content [62].

These threats are likely to carry forward into the 2026 local government elections. Allen and le Roux recommended that the IEC and the Government Communication and Information System (GCIS) hold a joint de-briefing to prepare better for the 2026 local government elections, and noted that some influencers had already been approached by external actors, which suggests that foreign influence operations may grow in future [60]. They do not,

however, predict how the threats will change. This study anticipates that because local government elections are contested at ward and municipal level, misinformation in 2026 will be more localised, more linguistically diverse and more dependent on community messaging groups. These are precisely the conditions under which the 2024 evidence shows monitoring to be weakest [1], [60].

4.3 The Multi-Stakeholder Electoral Information Landscape

Electoral information integrity in South Africa depends on several actors whose authority rests on public trust. The IEC manages elections under the Electoral Commission Act 51 of 1996, and the Electoral Act and its Electoral Code of Conduct bind registered political parties and candidates [59]. During the 2024 elections the IEC worked with other stakeholders on enhanced detection and fact-checking, on transparency and user education, and on implementing regulatory and policy frameworks [4]. This work included a Digital Disinformation Complaints process and the Real411 reporting platform, set up together with organisations including Media Monitoring Africa (MMA), through which anonymous submissions were assessed against the Electoral Code of Conduct and Section 89(2) of the Electoral Act [4].

The IEC also signed a non-binding Framework of Cooperation with Google, Meta, TikTok and MMA. Under it, the signatories committed to setting up a multi-stakeholder working group on disinformation and to working with the IEC and MMA on national initiatives against disinformation and for transparency in political advertising [4], [11]. The platforms also announced their own measures. Meta activated a South Africa-specific Elections Operations Centre, Google funded a fact-checking coalition with Africa Check and South African media, and TikTok set up a dedicated 'Mission Control' space and worked with the Code for Africa coalition on fact-checking; X issued no statement about the election [11]. The government also took part: a month before the election, GCIS set up a network of government computer scientists, fact-checking organisations, MMA and think tanks to identify and analyse reports of 'fake news' [60]. The South African National Editors' Forum (SANEF) set out informational risks for the election, including manipulated media and the hacking and impersonation of the IEC's social media presence [4]. Civil society organisations such as MMA, Africa Check and Defend Our Democracy formed part of the broader effort to maintain a safe and orderly electoral environment [4].

Each of these actors relies on trust that the information produced by the others is accurate and impartial, and the IEC itself was a primary target of disinformation. Leaders of the EFF and the MKP claimed that the IEC was behaving like a political party and favouring the ANC. Commissioner Janet Love was targeted as biased and corrupt, and social media posts claimed that the President had 'bought' the election from the IEC [4], [60]. The IEC was slow to counter false narratives, and by not explaining technical glitches promptly it left an information vacuum in which rumour and disinformation flourished [60]. Mare found that media coverage of the elections was overwhelmingly fair, but that some newspapers lent credibility to false claims questioning whether the IEC could be trusted [61]. The landscape is therefore multi-stakeholder not only because several organisations participate, but also because no single participant is trusted by all the others or by every segment of the electorate.

4.4 The South African Fact-Checking Ecosystem and Siloed Operations

The South African fact-checking landscape is dominated by Africa Check, which was launched in South Africa in 2012 and was the first independent non-profit fact-checker on the continent [7]. Africa Check operates outside conventional media outlets. It publishes its findings through digital platforms such as WhatsApp, TikTok and X, and depends on partners to translate its work into local languages and to broadcast it on radio and television [7]. Mainstream news organisations have also built their own capacity: News24 launched a full-time disinformation desk, and the Daily Maverick set up a Factcheck Hub with Africa Check [3]. During the 2024 elections, an election fact-checking coalition involving Africa Check, AFP Fact Check, the Daily Maverick, the Mail & Guardian, Caxton Media and SABC News, among others, tracked claims made by political parties and actors [4]. At the same time, Meta, Google and TikTok each maintained their own fact-checking partnerships [11], and Real411 provided a separate reporting channel [4].

These initiatives are valuable, but they are organised around individual organisations, platforms and funding arrangements rather than around shared verification infrastructure. A claim may be assessed by one organisation, labelled on one platform and reported through a separate complaints channel, and the outcome may not be visible to the other participants. Fact-checkers are not routinely present in South African newsrooms [60], and South African fact-checkers describe their work as resource-intensive and reactive: "When you wait for a claim to go viral and then you respond to it … much of the damage has already been done" [7]. In encrypted WhatsApp groups, Gondwe found that fact-checking efforts operated "in

parallel rather than in response to misinformation" [1]. Dependence on platform programmes also exposes the ecosystem to platform decisions. In January 2025, for example, Meta announced that it would end its third-party fact-checking programme in the United States and move to a Community Notes model [63]. That decision concerned the United States rather than South Africa, but it shows how quickly a platform can withdraw an arrangement on which fact-checkers depend.

International evidence shows why this fragmentation matters. Only 6.5% of the Snopes and PolitiFact articles analysed by Lee et al. concerned matching claims, which the authors interpret as evidence that fact-checkers have distinct ways of selecting and verifying claims. They conclude that fact-checkers need to collaborate and cross-check one another's findings so that the public receives the most reliable information [9]. One South African fact-checker described debunking the same false claims every two years, and Tatchou et al. note that such recurring "zombie claims" drain fact-checkers' resources [7]. Tatchou et al. do not draw this conclusion, but the recurrence of such claims suggests that a persistent, shared record of earlier verification outcomes would have practical value. None of the South African sources reviewed describes a common, cross-organisational record through which the IEC, fact-checkers, media organisations and civil society could see which claims have been assessed, by whom, on what evidence and with what outcome. Without such shared infrastructure, effort is duplicated, the public record is fragmented and consistency between organisations is difficult to demonstrate.

4.5 Structural Vulnerabilities in Fact-Checking Frameworks

Research on fact-checking has highlighted that organisations may differ in the claims they select, the evidence they consider and the procedures they use to evaluate information [8], [9]. Markowitz et al. compared The Washington Post's and PolitiFact's assessments of false and misleading statements by Donald Trump. They found 22.6% disagreement in statement selection and only moderate agreement in the scaling of deceptiveness, but nearly complete agreement on bottom-line veracity, and concluded that selection and scaling account for apparent discrepancies between fact-checkers [8]. Lee et al. found that 30.4% of 749 matching claims assessed by both Snopes and PolitiFact received diverging ratings. They attributed these differences to the granularity of the rating systems, the focus of each organisation, subtle differences in the information checked and the timing of the checks [9]. PolitiFact uses six rating levels, whereas Snopes adds more fine-grained categories such as

"Miscaptioned", "Scam" and "Satire" [9]. After these systematic differences were adjusted, only one conflicting verdict remained. Lee et al. also report earlier work finding that only 10% of the 2016 US presidential candidates' statements were checked by both The Fact Checker and PolitiFact, with lower agreement on statements in the ambiguous rating range [9].

These findings suggest that the main structural vulnerability is not that fact-checkers routinely reach opposite conclusions, but that their selection criteria and rating schemes are not standardised. As a result, their outputs are difficult to compare, aggregate or audit, and differences in selection can appear to an observer as biassed. Vinhas and Bastos interviewed 37 fact-checking experts from 35 organisations in 27 countries outside the Western, Educated, Industrialised, Rich and Democratic (WEIRD) world. They report that fact-checkers try to check a similar number of claims from each side of the partisan divide, a form of "bothsidesism" aimed at satisfying partisan social media audiences. Fact-checkers were also giving growing priority to viral content, and Meta's third-party fact-checking programme kept them from holding local politicians to account [64]. Tatchou et al. similarly identify a "neutrality bias" among South African fact-checkers, in which striving to appear even-handed can unintentionally obscure the truth [7].

The true/false verdict is itself a limitation. Monnier and Ségur argue that combating disinformation is not simply a matter of differentiating true from false information, and they note criticism that fact-checking favours verifiable facts at the expense of more complex situations [2]. Automated approaches inherit the same difficulty: in an evaluation of LLM-based fact-checking on PolitiFact claims, distinguishing between nuanced labels remained challenging [65]. A verification framework for a multi-stakeholder setting therefore needs to record more than a verdict. It should also record the claim selected, the rating scheme applied, the evidence relied on and the organisation responsible, so that methodological differences can be examined rather than inferred.

4.6 Public Trust, Polarisation and Allegations of Bias

Fact-checking depends on public trust, which is under strain in South Africa. In the 2025 Digital News Report, 55% of South African respondents said they trusted most news most of the time. This was well above the global average of 40%, but six percentage points lower than the 2022 high of 61%, and almost all news brands had experienced a decline in trust [3]. Worry about distinguishing real from fake news online was highest in Africa (73%), level

with the United States (73%), and 56% of South African respondents identified TikTok as a potential source of disinformation [3]. These figures need to be read with caution, however. The South African sample represents younger, English-speaking online news users rather than the national population [3], so it may not capture the experience of speakers of other South African languages, who are central to this study. More broadly, most research on how "fake news" affects democracies has been conducted in the Global North [59].

Fact-checkers are not immune to this erosion of trust. One South African fact-checker observed that "fact-checking is sort of becoming more and more seen as something that is partisan … a way of silencing people's freedom of expression", and another reported that people accuse fact-checkers of being liars when they debunk politically charged claims [7]. Research on online fact-checking services has identified trust and distrust as important considerations in the relationship between users and these services, with negative perceptions often grounded in concerns about integrity and perceived political bias [10]. Markowitz et al. note that people may question fact-checkers' credibility because they perceive them as biased, because of concerns around disinformation, or because of a general distrust of the media and institutions [8]. Monnier and Ségur similarly observe that some audiences regard the work of fact-checkers as further evidence that the mainstream media or the State is manipulating opinion [2].

Political polarisation makes corrections vulnerable to rejection because of their source. In the WhatsApp groups studied during the 2024 elections, corrections, including links to fact-checks and screenshots of news reports, were frequently distrusted, written off as propaganda or recast as foreign interference in South African politics [1]. Gondwe describes these groups as "epistemic enclaves", in which the group itself set the boundaries of truth and corrections were rejected because they originated outside its moral or political world [1]. Survey evidence from a United States panel of social media users also suggests that social media fact-checks are minimally effective: awareness of fact-checks, having one's own content fact-checked and even having content deleted did not deter users from sharing misinformation [66].

Trust is therefore not determined solely by the accuracy of an AI classifier or the professionalism of a fact-checker. The IEC's experience in 2024 suggests a parallel. As the body running the election, it was itself accused of rigging and of favouring the ANC [4], [60], and by analogy a single authority that issues verdicts on contested political claims is likely to

attract similar allegations of bias. Brandtzaeg and Følstad argue that fact-checking services must be transparent about their working process and could act as “platforms for collaborative verification” [10], and in an experiment with students, groups of three identified false posts more accurately than individuals, though no better than a simple majority of independent judges [67]. By analogy, verdicts should come from several independent organisations rather than one. A monitoring system also needs to provide a means through which verification decisions can be examined and traced. Stakeholders require mechanisms for recording who performed a verification, what decision was reached, what evidence it relied on and when that decision was made. Transparency of this kind will not persuade every sceptic, but it allows criticism to be directed at verifiable decisions rather than at the presumed motives of an institution. A monitoring system that produces classifications without maintaining an auditable record may therefore address the detection problem without addressing the transparency and accountability requirements of a polarised, multi-stakeholder environment.

4.7 Artificial Intelligence and Misinformation Detection

Artificial intelligence (AI) and machine learning have increasingly been applied to misinformation and fake-news detection. Early approaches commonly relied on manually designed features or statistical representations of text, followed by supervised classification algorithms such as Logistic Regression, Support Vector Machines and Naive Bayes [12]. One commonly used representation is Term Frequency-Inverse Document Frequency (TF-IDF), which represents text according to the frequency and relative importance of words within a collection of documents. Such feature-based methods provide useful classification baselines because they are comparatively simple to train and interpret, but they have limited ability to represent relationships between words and the broader context of a statement.

Deep learning approaches subsequently enabled models to learn more complex representations directly from textual data. Transformer-based language models are particularly relevant because they use attention mechanisms to represent relationships between tokens within a sequence. Transformer-based approaches have become an important direction in misinformation detection research because they can capture contextual information that traditional feature-based approaches may not represent effectively [12], [26]. In the systematic review by Rananga et al., approximately 81% of the studies preferred deep learning over traditional machine learning, and research has examined content-based, social context-based and hybrid approaches to detection [12].

Large language models (LLMs) extend these approaches. Fine-tuned BERT-like encoders generally outperform decoder-only LLMs on classification, whereas LLMs are more robust to text perturbations [26]. Sahitaj et al. evaluated Llama-3 models with 3, 8 and 70 billion parameters on 17,856 PolitiFact claims, using evidence retrieved through restricted web searches and combining verdict classification with generated explanations. Larger models performed better, evidence integration improved performance across all models, and distinguishing between nuanced labels remained difficult [65]. Together, these studies point towards a pipeline of claim detection, evidence retrieval, verdict classification and explanation, rather than single-step classification.

However, misinformation detection is not equivalent to determining objective truth. A claim may require information about its source, publication time, surrounding events, supporting evidence and intended meaning before its factual status can be established [2], [65]. Generative models can also produce convincing output that is unsupported or hallucinated, which reinforces the need for verification and explainability mechanisms [62], and explainability in automated fact verification lags behind the wider field of explainable AI [68]. These limitations are particularly important in politically sensitive and multilingual applications. In Gondwe's study of the 2024 elections, the AI-powered verification tool used had not been peer-reviewed for African languages, so its classifications were cross-checked and corrected by bilingual human coders [1]. An AI classifier can therefore be used to identify potentially misleading content and prioritise material for further investigation, but its output should not automatically be treated as a final factual judgement. This distinction provides an important basis for combining automated detection with subsequent verification.

4.8 Multilingual and Low-Resource Natural Language Processing

A major challenge in applying NLP to misinformation monitoring is linguistic diversity. NLP research has historically concentrated on languages for which large quantities of digitised text, annotated datasets, linguistic resources and pretrained models are available. In contrast, many languages have fewer computational resources and are therefore commonly described as low-resource languages [12]. Low-resource status is multidimensional and may involve limitations in labelled data, linguistic resources, benchmarks, pretrained models and tools tailored to the language [12]. In the review by Rananga et al., English datasets were used in 67% of misinformation detection studies, followed by Arabic (14%) and Chinese (11%), and fewer than 10% of studies focused on low-resource languages [12].

This issue is particularly relevant to African NLP. Africa contains substantial linguistic diversity, but this diversity is not proportionally represented in many state-of-the-art NLP systems [12]. South Africa has 12 official languages, yet the Institute for Strategic Dialogue found that platforms moderate non-English content inadequately [11]. In 2024, Meta claimed to have "the largest fact-checking network of any platform", covering English, Afrikaans, isiZulu, Sesotho and Setswana, which are five of South Africa's 12 official languages [11]. Research on South African languages has consequently investigated approaches for misinformation detection in languages such as Tshivenda, Sesotho and Sepedi [13], [14], [15]. This work is still at an early stage. The Tshivenda study, for example, reported the same accuracy on English and Tshivenda data, but its dataset was too small to calculate F1 or recall [13]. These studies therefore do not yet show how well models developed for high-resource languages perform on South African languages, and Rananga et al. identify "a significant lack of validated models specifically designed for African languages" [12].

Data availability is an important component of this problem. Supervised misinformation classification requires labelled examples from the target domain and language. When only small quantities of labelled data are available, models may have difficulty learning language-specific patterns and contextual distinctions. The Tshivenda, Sesotho and Sepedi studies each relied on English datasets translated into the target language because native labelled data were scarce [13], [14], [15], and the Sesotho study found that translation quality had a substantial impact on model performance [14]. Translated datasets may also fail to capture how misinformation is actually expressed by speakers of these languages. Data augmentation has therefore been investigated as a possible approach for improving classification in low-resource settings [15], and pretrained transformer models have been applied to fake-news detection in other low-resource languages such as Bangla [69].

4.9 Linguistic Complexity: Code-Switching, Vernacular and Audio Content

South African social media discourse frequently combines languages within a single utterance. In multilingual societies like South Africa, speakers often move between English and their home languages mid-sentence, which complicates annotation [70]. In a Twitter corpus covering Sepedi, Sesotho, Setswana, isiXhosa and isiZulu, approximately 30% of the tweets were code-switched with English [24]. Rananga et al. identify the prevalence of code-switching, the lack of standardised writing systems and limited annotated datasets as factors that make misinformation detection harder for African languages [12].

Code-switching and informal language also affect model performance. Mabokela et al. found that code-switching posed additional challenges for pre-trained language models in the Sotho-Tswana languages, and that tweets expressing sarcasm or mixed emotions were difficult even to label accurately [70]. A related evaluation found zero-shot GPT-4 far weaker than a fine-tuned Afrocentric model on such text [24]; although these studies concern sentiment rather than misinformation, general-purpose LLMs cannot be assumed to handle code-switched South African text reliably.

Vernacular slang and localised expressions also create gaps in platform moderation. The Institute for Strategic Dialogue found that TikTok took no action against content that used non-English South African vernacular to target and degrade women, even though the equivalent English terms were banned on the platform [11]. Of 57 potentially violating TikTok comments aimed at women politicians, 33 contained non-English words, and 13 switched into another language for their most harmful content [11]. As noted in Section 4.1, senior editors worried that WhatsApp groups in vernacular languages may have done more to spread false information during the 2024 elections than open platforms [60].

Misinformation also travels as audio and video. Deepfake audio, voice-cloned speech and decontextualised recordings were all documented during the 2024 elections [1], [60], yet most detection research is text-based: approximately 73% of the studies reviewed by Rananga et al. used text data and only 2% examined speech or audio [12]. A detection component for South Africa therefore needs to account for code-switched, informal and multimodal content rather than assuming clean, monolingual text.

4.10 Transformer-Based Models for African Languages

The development of multilingual language models has created new opportunities for applying transformer-based NLP to African languages. Multilingual models can transfer representations learned from languages with greater computational resources to languages with fewer resources. This makes transfer learning particularly relevant where large language-specific datasets are unavailable [12].

Research on African NLP has expanded, with datasets and transformer-based models developed for a growing number of African languages, although much of this work has concentrated on West and East African languages while Southern African languages have received less attention [70]. Some recent multilingual models now cover South African languages. Mabokela et al. note that AfroXLM-R, adapted to 17 African languages, covers

isiZulu, isiXhosa and Sesotho; that SERENGETI covers 517 African languages, including all five South African languages they examined; and that PuoBERTa is a monolingual Setswana model. Other African models, such as AfriBERTa, which was trained on 11 African languages, cover none of those five languages [70]. The Zabantu pre-trained models have also been applied to Sepedi misinformation classification [15]. Such developments provide a foundation for investigating whether transformer-based models can support misinformation classification across languages with different levels of available training data.

The use of transformer models does not, however, remove the need for empirical evaluation. Different model architectures and pretrained language resources may respond differently to classification tasks, and performance in one language does not necessarily predict performance in another. For example, AfroXLM-R performed better on isiXhosa and isiZulu than on Sepedi, Setswana and Sesotho, a gap the authors attribute to the greater amount of pre-training data available for isiXhosa and isiZulu [70]; their ensemble reached average weighted F1-scores above 77% for the Nguni languages, which include isiZulu, but above 63% for the Sotho-Tswana group, which includes Sepedi [70]. Consequently, multilingual misinformation detection requires evaluation across the individual languages represented in the application domain rather than relying solely on an aggregate performance measure.

4.11 Single Points of Failure in Centralised Record-Keeping

Most verification outcomes are currently held in systems controlled by individual organisations, such as a fact-checker's website, a platform's labelling system or a complaints portal operated by a single entity. Centralised trust models concentrate control and can create single points of failure. Nair et al. argue that the typical centralised cloud trust model is costly to maintain, prone to network congestion and vulnerable to a single point of failure, and that because it lacks openness and traceability its trust ratings are not universally accepted [71]. Wang et al. note that storing news data on cloud servers takes physical control of the data away from its owners, creating a risk that fake news will be spread and genuine news altered [72]. The authors of ChainSEAL, a prototype blockchain evidence ledger presented as a short poster paper, likewise describe traditional centralised off-chain storage as insecure, poorly traceable and open to tampering [73].

The 2024 elections showed how failures in centralised systems can feed misinformation. After the IEC's results board and website crashed and stayed offline for more than an hour, allegations that the IEC's delays and counting problems were deliberate gained traction, and

the MKP claimed that an IT entity had interfered and was “rigging the system” [4]. Media coverage then focused on how well the results dashboard worked after the crash [61]. Election observers also noted malfunctions in the deployment of voter management devices [74], and IEC technical glitches more broadly were framed as malicious attempts to sway the election [60]. The outage did not demonstrate tampering, and the results were not held by the IEC alone. Results were announced publicly and posted outside each polling station, party and candidate agents received a copy, and observers commended these measures for contributing to transparency [74]. Parties also lodged 579 formal objections with the IEC, including a petition calling for a recount [4]. The episode therefore shows both how quickly suspicion fills the gap when a centrally held record becomes unavailable or opaque, and the value of copies held independently of the custodian.

Fact-checking and monitoring records have no equivalent of the results slips given to party agents. A verification record held by one organisation can be edited, removed or made unavailable through error, technical failure, external pressure or partisan interference, without other stakeholders being able to detect the change. South Africa’s history of apartheid-era censorship makes this concern particularly sensitive, and Thukuse cautions that measures against misinformation must not themselves endanger the freedoms on which democracy depends [75]. In a polarised environment, even a legitimate correction by a single custodian may be perceived as tampering. These are structural rather than organisational weaknesses, because they arise from where control over the record sits rather than from the integrity of any particular organisation. Blockchain systems, by contrast, are designed for decentralised record-keeping in which no single entity can change the data on its own [76]. In a permissioned blockchain, every authorised party sees the same data, and no party can alter it without the agreement of the other members [17].

4.12 Blockchain for Transparency, Traceability and Provenance

Blockchain technology provides a distributed approach to maintaining records between multiple participating entities. Rather than relying exclusively on a central database controlled by a single organisation, blockchain systems maintain a shared ledger in which transactions are recorded and subsequently replicated or validated according to the rules of the network [77]. Cryptographic mechanisms can provide evidence of whether recorded information has subsequently been modified. Because information on a blockchain must be validated by several independent entities, Lumineau et al., drawing on Brown, argue that

blockchains allow parties who do not fully trust one another to agree on a common set of facts [17].

Blockchain has previously been proposed for addressing information disorder and the traceability of news [78], for fake-news traceability and verification [72] and for establishing the origins of online content [49]; these systems are reviewed in Section 5.3.

The principal relevance of blockchain to the present research is therefore provenance and auditability, rather than classification. Blockchain cannot independently determine whether a political statement is true or false. Lumineau et al., drawing on Catalini, note that the immutability of blockchain information is useful only when the information originally entered is accurate. Drawing on Halaburda, they add that information which is not native to the blockchain gives rise to a "first mile/last mile" problem, in which verifiers must evaluate external information before it is recorded [17]. They also describe the "oracle problem", in which "flawed or incorrect information provided by the transacting parties inappropriately triggers the execution of a smart contract", and argue that where transactions are hard to verify, human intervention becomes necessary [17]. The validity of information entered into a ledger therefore still depends on the processes and entities responsible for producing it. Blockchain can instead preserve a record of a verification event, including information such as an identifier, timestamp, verification status and cryptographic reference to associated material [49], [72], [79].

This distinction is important when integrating blockchain with AI. AI can support the identification of potentially misleading content, while blockchain can preserve information about subsequent verification activities. The two technologies therefore address different aspects of the misinformation-monitoring problem rather than performing the same function.

4.13 Permissioned Blockchain and Multi-Stakeholder Verification

Blockchain networks can be implemented using different governance and participation models. Permissionless blockchains generally allow participants to interact with the network without requiring membership to be established by a central authority, whereas permissioned blockchain networks restrict participation to identified and authorised entities. Permissioned approaches are particularly relevant to applications involving organisations that need controlled access, identifiable participants and defined governance procedures [77]. Governments adopting blockchain often opt for permissioned networks precisely because

they retain greater control and can more readily comply with legal constraints on data handling [76].

Hyperledger Fabric is a permissioned blockchain framework designed for distributed applications involving known participants. Hasnain et al. identify several Fabric features that support confidentiality between organisations: digital certificates that establish the legitimacy of participating organisations, separate channels for information distribution, isolation of information between organisations on the same channel, and the storage of transaction details apart from the on-chain record [80]. These characteristics make permissioned blockchain relevant to a consortium in which the IEC, fact-checking organisations, media organisations and civil society monitors could participate in a common verification process.

A consortium model can also provide a basis for distributing responsibility for verification. Rather than allowing one organisation to control the complete verification record, participating organisations can operate according to predefined policies governing which entities may submit, review or finalise records. Lumineau et al.'s example of IBM, AIG and Standard Chartered collaborating on a blockchain-based multinational insurance policy [17] is a close analogue: its members must cooperate, but none should own the shared record. Such an arrangement is consistent with the use of blockchain for provenance and institutional accountability, while avoiding the assumption that decentralisation alone guarantees the correctness of a verification decision. Governance remains essential. Drawing on Halaburda, Lumineau et al. observe that in permissioned blockchains, trust between participants remains a central mechanism for reducing opportunism and managing the first mile/last mile problem [17].

4.14 Regulatory Constraints under POPIA

Any system that records claims, evidence and verification activity in South Africa must comply with the Protection of Personal Information Act 4 of 2013 (POPIA), enacted in November 2013 [20], [81]. The Act's main provisions came into effect on 1 July 2020 [82]. POPIA regulates the processing of personal information through eight conditions for lawful processing: accountability, processing limitation, purpose specification, further processing limitation, information quality, openness, security safeguards and data subject participation [20], [81].

Several provisions are directly relevant to a misinformation monitoring system. Processing must be justified, for example by the data subject's consent, by the responsible party's

legitimate interests or, for a public body such as the IEC, on the ground that it is "necessary for the proper performance of a public law duty"; data subjects may object to processing that relies on legitimate interests or a public law duty (section 11) [20]. Section 14 provides that "records of personal information must not be retained any longer than is necessary for achieving the purpose for which the information was collected or subsequently processed", unless an exception applies, such as retention required by law or for historical, statistical or research purposes with appropriate safeguards [20]. Once retention is no longer authorised, records must be destroyed, deleted or de-identified as soon as reasonably practicable, "in a manner that prevents its reconstruction in an intelligible form" [20]. Data subjects may request the correction or deletion of personal information that is "inaccurate, irrelevant, excessive, out of date, incomplete, misleading or obtained unlawfully" (section 24) [20], [81]. Responsible parties must also "secure the integrity and confidentiality of personal information" (section 19), and information that has been "de-identified to the extent that it cannot be re-identified" falls outside the Act (section 6) [20], [81].

Election misinformation records are likely to contain personal information, including the names and social media identities of people who made or shared claims, screenshots of posts and messages, and information about individuals named in claims. Political persuasion is classified as special personal information, as is information about a person's alleged commission of an offence (section 26) [20], [81]. This matters because election disinformation frequently alleges criminal conduct such as bribery, vote buying and rigging [4]. Processing special personal information is prohibited unless an authorisation applies, for example the data subject's consent, information deliberately made public by the data subject, or historical, statistical or research purposes that serve a public interest with sufficient privacy guarantees (section 27) [20]. The specific authorisation concerning political persuasion applies to institutions founded on political principles, such as political parties (section 31), rather than to independent monitors [20]. POPIA excludes processing solely for journalistic purposes to the extent that the exclusion is necessary "to reconcile, as a matter of public interest, the right to privacy with the right to freedom of expression" (section 7) [20]. Where a journalist is bound by a code of ethics that provides adequate safeguards, that code applies instead of the Act (section 7(2)) [20]. It is not clear, however, that this exclusion would extend to a shared monitoring infrastructure in which the IEC, civil society organisations and technical operators participate alongside journalists. POPIA also operates within the constitutional protection of freedom of expression under Section 16 of the Constitution,

which does not exclude false statements from protection. Any law that restricts the spread of information, true or false, must therefore be justified under the limitation clause [59].

4.15 The Immutability–POPIA Paradox

The value of a blockchain lies in its permanence. Records are append-only, replicated across participants and cannot be altered unilaterally [17], [76]. POPIA, by contrast, requires that personal information be corrected when inaccurate, deleted when its retention is no longer authorised, and destroyed in a manner that prevents its reconstruction (sections 14, 16 and 24) [20]. This creates a tension between a ledger's "right to remember" and the data subject's right to have information corrected or removed. Finck, analysing the same tension under the European Union's General Data Protection Regulation (GDPR), observes that blockchains are usually designed deliberately to make the unilateral modification of data difficult or impossible, which is hard to reconcile with requirements that personal data be amended and erased in specific circumstances [83]. Almiani et al. similarly note that the "right to be forgotten" is at odds with blockchain's append-only design, and report that this tension has led many implementations to favour permissioned or hybrid models [76]. A South African public-sector study likewise notes the concern, raised by Dowelani et al., that transparent identities on a ledger could compromise POPIA [84].

The tension is not limited to deletion. Fact-checkers correct their work: South African fact-checkers report that they retract or edit an article when good-faith criticism identifies an error [7]. Because immutability is useful only when the original information is accurate, as Lumineau et al. note in drawing on Catalini [17], a verification ledger must support correction without erasing the history that gives it evidential value. An appended correction can supersede an earlier record, but the earlier record remains on the ledger. This may satisfy auditability while failing POPIA's deletion requirement if the original record contained personal information.

Replication intensifies the problem. A blockchain ledger is replicated or validated across the network [77], so personal information written on-chain may be copied to several participating organisations and cannot be destroyed by any one of them [83]. The design question is therefore not whether to choose immutability or privacy, but which data are recorded immutably. The approaches discussed in the next section address this question by keeping personal information off the ledger and recording only what is required to prove the integrity and history of a verification decision.

4.16 Off-Chain Storage, Hashing and Cryptographic Privacy

A further consideration in blockchain-based information systems is the distinction between data that must be permanently recorded on the ledger and data that can be maintained outside the blockchain. Storing large documents, evidence files or potentially sensitive information directly on a blockchain can introduce storage, privacy and scalability concerns. Research on blockchain-based information systems has therefore investigated hybrid architectures in which substantive data are maintained off-chain while integrity-related information is recorded on-chain [38], [48].

Cryptographic hashing provides one mechanism for connecting off-chain information with an on-chain record. A cryptographic hash produces a fixed-length representation of input data. If the underlying content changes, the resulting hash is expected to change, allowing a later verification process to determine whether the content corresponds to the previously recorded representation. Provenance research has investigated blockchain-based timestamping and tamper-evident records for establishing the history and integrity of digital information [79]. Examples include the ChainSEAL evidence ledger [73] and Wang et al.'s design for news data [72], both reviewed in Section 5.

This approach is relevant to misinformation monitoring because claims and supporting evidence may contain information that should not be permanently exposed to every participant in a shared ledger. A system can instead maintain the substantive material outside the blockchain while recording a cryptographic reference and relevant metadata on the ledger. The resulting structure separates content storage from verification provenance, allowing the blockchain to provide an auditable record without requiring the complete information object to be stored on-chain [38], [48]. Finck recommends this approach under the GDPR, noting that personal data stored off-chain and linked to the ledger only by a hash can be rectified and erased [83]. In relation to POPIA, this means that personal information held only off-chain could be corrected or deleted there, while the ledger continues to show that a verification event took place.

Permissioned platforms and newer cryptographic techniques offer further controls. Hyperledger Fabric's channels can restrict which organisations see which transactions, and its Identity Mixer (Idemix), described as a zero-knowledge proof technology, provides privacy-preserving features such as anonymity, although with technical limitations [80]. More generally, Almiani et al. recommend that zero-knowledge proofs, homomorphic encryption,

secure multiparty computation and similar privacy-preserving methods be adapted for public-sector use and tested for legal compliance as well as technical performance [76]. In principle, such techniques could allow a participant to demonstrate that a verification met an agreed rule, such as endorsement by a required number of organisations, without revealing the underlying personal information.

These techniques reduce, but do not remove, the compliance question. Almiani et al. caution that blockchain metadata can reveal information even when hashing and encryption are applied [76]. Whether a hash derived from personal information counts as de-identified information under section 6 of POPIA depends on whether it can be re-identified [20]. Finck identifies the same open question under the GDPR: when off-chain personal data is erased, the hash that remains on the ledger may itself still be personal data [83]. A hash of a short or predictable value, such as a phone number, could potentially be re-identified by hashing candidate values and comparing the results. A POPIA-compliant design must therefore treat on-chain metadata as potentially personal and keep it to a minimum.

4.17 Integration of Artificial Intelligence and Blockchain

The preceding discussion demonstrates that AI and blockchain address different requirements within misinformation monitoring. Machine learning provides mechanisms for processing and classifying large volumes of information, while blockchain provides mechanisms for recording events and maintaining provenance across multiple participating entities. Research examining the integration of AI and blockchain has identified these complementary capabilities. Kuznetsov et al. note that storing data and AI decisions on a blockchain can create a secure and transparent record of an AI system's decision-making that can be audited and scrutinised, addressing concerns about the "black box" nature of deep learning, and that blockchain can help ensure the provenance and integrity of data used by AI models [85].

An integrated misinformation-monitoring system can therefore be conceptualised as a sequence of complementary activities. First, AI can analyse incoming content and identify claims that require attention. Second, evidence and contextual information can support subsequent assessment. Third, authorised human or institutional participants can perform verification. Finally, relevant information about the verification event can be recorded in a tamper-evident ledger. This separation of responsibilities is important because it prevents the blockchain component from being interpreted as a mechanism for establishing the

truthfulness of the AI output, and because human verification addresses the first mile/last mile problem before information is committed to the ledger [17].

For a multilingual South African election environment, the integration also needs to account for linguistic diversity, low-resource languages, institutional participation, information integrity and data protection. Existing research has addressed these aspects individually, as the preceding sections show, but not as a complete framework combining multilingual AI-based detection with a shared, auditable and privacy-compliant verification record.

4.18 Background to the Present Research

The 2024 elections demonstrated that election misinformation in South Africa is now networked, algorithmically amplified, partly hidden in encrypted groups, increasingly synthetic and frequently directed at the IEC itself [1], [4], [60], [61], and, as argued in Section 4.2, the 2026 local government elections are likely to face these threats in a more localised form. The literature reviewed above therefore establishes five closely related requirements for an election misinformation monitoring system.

First, the system must be capable of processing multilingual content in a language environment where labelled resources are limited [12], [13], [14], [15], [24], [69], [70]. Second, automated classification should support rather than replace contextual verification, because classification alone cannot establish the factual status of every political claim, and because fact-checking organisations differ in how they select and rate claims [2], [8], [9], [65]. Third, verification records must be shared across organisations that do not fully trust one another, and they must not depend on a single custodian whose failure or alleged partiality could undermine the record [4], [7], [17], [71]. Fourth, the verification process must be transparent and traceable, so that allegations of bias can be tested against a record of who verified what, using which evidence and when [1], [10], [49], [72], [78], [79]. Fifth, the system must comply with POPIA, which requires that personal information can be corrected and deleted; the conflict with ledger immutability must therefore be resolved through the system's architecture rather than ignored [20], [76], [83].

These requirements motivate an integrated approach in which artificial intelligence and consortium blockchain perform complementary functions. The AI component addresses the analytical challenge of identifying potentially misleading multilingual content. The permissioned blockchain component addresses the institutional challenge of preserving verification and provenance information across competing stakeholders. Off-chain storage

combined with on-chain cryptographic references reconciles that record with POPIA. The combination consequently provides the conceptual foundation for investigating an integrated framework for transparent, tamper-resistant, privacy-compliant and multilingual election misinformation monitoring in the South African context.

5 Related Work

This study draws on two bodies of work that have so far developed separately: the automatic detection of misinformation in South Africa's lower-resourced languages, and the use of permissioned blockchains to keep tamper-evident records shared by institutions that do not fully trust one another. A smaller third body of work proposes shared infrastructure, including blockchains, specifically for fact-checks and fake news. Sections 5.1 to 5.3 review each in turn.

5.1 Misinformation Detection in Low-Resource South African Languages

5.1.1 South African detection studies built on translated data

Rananga et al.'s review of misinformation detection (Section 4.8) calls for new datasets in low-resource languages, for data augmentation and transfer learning, and for fine-tuning pre-trained multilingual transformers such as XLM-R [12]. Recent South African studies have acted on these recommendations, each overcoming the lack of a labelled corpus in the target language by machine-translating an existing English dataset.

Mokoena et al. [14] translated the CoAID COVID-19 dataset into Sesotho with two translation systems and found that the same SVM classifier scored an F1 of 0.908 on the better translation against 0.866 on the other, concluding that translation quality has a direct, measurable effect on detection performance. Manaswe et al. [15] translated 1,500 sentences of CoVID19-FNIR into Sepedi and tested six data augmentation techniques: classical models scored highly on the augmented data, while the fine-tuned Zabantu encoder reached at most 79% accuracy, so transformer performance on such limited data remains modest. Sethosa et al. [16] applied the same strategy to hate speech in isiZulu with TF-IDF classifiers, but found that translation tools struggled to render offensive text and cultural context accurately, and recommend transformer models for future work.

De Jager et al. [86] addressed local context. They introduced the first South African misinformation dataset, built from complaints submitted to the Real411 platform, and found that adding South African samples to a multimodal BERT and ResNet model significantly improved detection in the South African setting, although Real411 is predominantly English.

These studies show that detection in Sesotho, Sepedi and isiZulu is feasible at useful accuracy without native corpora, that machine translation of a labelled English source is the accepted way to obtain training data in these languages and that its quality matters, and that

locally relevant content improves detection. None addresses elections, however, and most of their models are classical TF-IDF classifiers. This study keeps the translation step but changes its source (Section 3.1), and retains TF-IDF baselines for comparison with these studies (Section 3.2).

5.1.2 Language models for African-language classification

Mabokela et al. [24] found that fine-tuned AfroXLMR outperformed generative LLMs such as GPT-4 and BLOOM on sentiment classification of tweets in five South African languages, including Sepedi and isiZulu, and attribute its advantage to pre-training on African-language text that copes with agglutinative morphology and English code-switching. Continued pre-training of AfroXLM-R on African social media text improved F1 by between 1% and 30% across 19 languages, although its corpus included isiZulu and not Sepedi [25]. For English fake news, Raza et al. [26] found that fine-tuned BERT-like encoders outperformed fine-tuned decoder-only LLMs on every classification metric at lower cost, while the LLMs were more robust to perturbed text, and that AI-generated labels checked by people produced better classifiers than labels inferred from the news source. Zhang et al. [87] likewise found that encoders generally outperform LLMs on classification with fewer computational resources, although without fine-tuning the LLMs, which did better on knowledge-intensive tasks such as hallucination detection, arguably the task type closest to fact verification. These comparisons inform the choice of models in Section 3.2.

5.1.3 LLM-generated training data

Nadăş et al. [21] survey LLM-generated training data for settings where labelled data is scarce, expensive or sensitive; one study they review found gains of 3–26% in accuracy or F1-score when 100 synthetic samples were added to 100 real ones. They recommend topic-controlled or randomised prompting to increase diversity, and filtering and human validation because generated text can contain factual errors, may not match the style and distribution of real data, and can amplify bias. The prompt axes, back-translation check and stated limitations in Section 3.1 follow these recommendations, although the generated labels were not individually verified by people.

5.1.4 Parameter-efficient fine-tuning

Shuttleworth et al. [28] examined whether Low-Rank Adaptation (LoRA) and full fine-tuning learn equivalent solutions. Even when both reached the same task accuracy, their weight matrices differed: LoRA introduced new high-ranking singular vectors, which the authors call

intruder dimensions, that made the model forget more of its pre-training distribution and accumulated when tasks were learned in sequence. Setting the LoRA scaling factor to twice the rank ($\alpha = 2r$) produced fewer intruder dimensions and better generalisation than a smaller α. Section 3.2 therefore compares the two methods directly rather than assuming they are equivalent, and sets $\alpha = 2r$.

5.2 Permissioned Blockchains for Shared Institutional Records

The blockchain layer of this study records the decisions of organisations that do not fully trust each other, keeps documents off-chain and gives organisations different rights. Studies in health care, criminal justice, public administration and supply chains, where institutions also share records that may later be disputed, have established most of the design patterns it adopts.

5.2.1 Health records

Tith et al. [88] built a Hyperledger Fabric consortium through which hospitals share electronic health records: member institutions' certificate authorities issue the identities, the ledger holds only the address and hash of each record, the records stay in each hospital's existing system, and every access is logged for auditing, showing that a consortium ledger can sit on top of institutions' existing systems without replacing them. Hasnain et al. [80] reviewed 57 studies of Fabric in health care and found that it was chosen mainly for privacy, security, integrity, traceability and availability, was often combined with the InterPlanetary File System (IPFS) and benchmarked with Hyperledger Caliper, but had seldom been evaluated under real-world conditions.

5.2.2 Criminal justice and evidence management

Criminal justice is the closest institutional analogue to election misinformation reporting: records must be shared among agencies with separate mandates and must withstand later scrutiny. Ramazhamba and Venter [89] proposed ShareCrimE, a Fabric network with Raft ordering for sharing case data among six South African criminal justice entities, each with its own certificate authority, to counter the loss and theft of case dockets. It is the nearest South African precedent for a multi-institution Fabric consortium, but it stored the case data itself in the ledger's world state and was evaluated only through a functional simulation. Amin et al. [90] kept sensitive crime-report data in Fabric private data collections and showed with Caliper that Fabric can handle reporting workloads at sub-second latency. Tasnim et al.'s

CRAB [91] lets law enforcement agencies and courts add criminal records on a permissioned Ethereum network, while approved organisations such as airports can only look them up.

Mahajan et al. [73] presented ChainSEAL, an evidence ledger that stores encrypted evidence files on IPFS and anchors only the SHA-256 hash, the IPFS content identifier (CID) and chain-of-custody metadata on an Ethereum-based chain, with smart-contract roles for courts, police officers, forensic investigators and lawyers. They show that storing a 1 MB file fully on-chain would need about 724 million gas, far above Ethereum's block gas limit of about 30 million, and that changing a single bit of a file changes more than 100 bits of its SHA-256 hash, so tampering is detected when the hash is recomputed.

5.2.3 Government records and document integrity

Alotaibi et al. [92] used design science research, the approach this study follows, to build GovBlockchain, a conceptual model for transparent government procurement records in which an auditor can view all documents and citizens only non-sensitive details. They chose Hyperledger Fabric over Ethereum for its permissioned membership, private channels and pluggable consensus, making the study a direct methodological precedent for using Fabric to open institutional decisions to independent audit. Mishra and Ganesan [93] and Rashmi and Harish [94] anchored SHA-256 hashes of e-government documents and academic credentials on Ethereum-based networks, showing that the hash of an off-chain document is enough to verify it later.

5.2.4 Off-chain storage and access control

Zhu et al. [48] formalised the on-chain/off-chain pattern in IPFS-BC, which keeps raw files in IPFS and only their CIDs and metadata on a consortium blockchain, using about 75% less storage than storing the files on-chain; they note that off-chain nodes may be unavailable, off-chain data is weakly protected and access control is hard to unify across systems. Dai et al. [95] checked permissions inside Fabric chaincode against roles recorded in the ledger state and still reached 214 transactions per second, but their geographically split deployment showed how strongly node placement affects latency, which is why Section 3.7 reports the benchmark hardware.

5.2.5 Implications for this study

Across these domains a common design emerges, and this study adopts it: a consortium of known institutions, each issuing identities through its own certificate authority; Raft ordering; documents kept off-chain, with only their hash or content identifier on the ledger; separate

roles for submitting, deciding and reading; and Hyperledger Caliper for measuring performance. Where these systems enforce roles on the ledger, however, the roles are bound to Ethereum account addresses [73], [93] or to role records in the ledger state [95] rather than to certificate attributes that every endorsing peer can check (Section 3.4.2), and read-only access for external parties is conceptual [92] or enforced outside the ledger [91].

5.3 Shared Fact-Checking Infrastructure and Blockchain Approaches

Shared infrastructure for fact-checks already exists outside the research literature. ClaimReview markup lets fact-checkers tag their articles so that tools such as Google Fact Check Explorer can index fact-checks from around the world in one searchable place [96], and the International Fact-Checking Network's CoronaVirusFacts/DatosCoronaVirus Alliance pooled fact-checks from organisations in more than 70 countries in a shared database during the COVID-19 pandemic [97]. These resources make published verdicts easier to find, but they collect verdicts that each organisation reaches on its own; they are not tamper-evident, and they do not produce joint verdicts. The studies below use blockchains to make such records tamper-evident.

Huckle and White [49] built Provenator, an Ethereum application that records provenance metadata for digital media so that anyone can check a resource's hash against its creator's record, motivated by a photograph of ballot boxes from a United Kingdom election reused in the 2016 United States election as supposed evidence of vote rigging. A hash matches only an identical file, so changing a single pixel defeats the tool, and, the authors argue, technology can verify the origin of a resource but not the truth of a story, which "takes human skills". This study accepts both points: its ledger proves the integrity of a report, not its truth, and human fact-checkers make the verdict on each claim.

Qayyum et al. [78] proposed, but did not implement, a framework in which smart contracts enrol and revoke news publishers with reputation scores and a Merkle tree lets anyone check whether a news item is on the chain; they propose semantic similarity measured with natural language processing to compare reports, and name AI-based assessment of news trustworthiness as future work. Wang et al. [72] kept full news data off-chain in a cloud database, with abstracts and polynomial commitments on-chain kept consistent by an updatable commitment scheme that was analysed formally but not evaluated empirically.

Farhan et al. [38] proposed VeriTrust, the design closest to this study, which combines self-sovereign identity, Merkle-tree anchoring of content-metadata hashes on an

Ethereum-compatible chain, off-chain storage on IPFS, AI flagging of suspicious content, voting by fact-checkers whose expertise is attested by credentials, and a reputation system, with no personal data on-chain. It thus sets out the sequence that this study implements: automated flagging, human verification and a tamper-evident record. However, it is explicitly conceptual, with performance figures projected from published benchmarks; its token rewards and optional stake-based voting suit an open network rather than a consortium of known institutions; and its authors acknowledge that gas costs may become a bottleneck and that ledger immutability conflicts with erasure rights, the concerns that Sections 3.4.1 and 3.5.1 address.

Kuznetsov et al. [85] note that a ledger can record the provenance of training data and every version of a deployed model, but that combining AI and blockchain also brings risks, from adversarial attacks and bias to smart contract vulnerabilities, key management and the scalability trilemma. This study records the model version string with each report (Section 3.5.1), so that every verdict can be traced to the model that produced the initial label.

These studies show that a ledger can make decisions about misinformation tamper-evident and auditable, and VeriTrust shows how automated detection and human verification can be organised around one. None, however, has been implemented and benchmarked end to end, none is designed for a permissioned consortium of known institutions with certificate-based identities, and none considers African languages, elections in Africa or a specific data protection law such as POPIA.

5.4 Summary and Research Gap

Taken together, the reviewed work makes this research feasible: translated training data, Afrocentric encoders, LLM-generated corpora and Fabric consortia with Raft ordering, off-chain storage and hash anchoring have each been demonstrated. No study, however, brings these elements together. In particular:

(i) No labelled election-misinformation corpus or detector exists for Sepedi or isiZulu. The South African detection studies address COVID-19, hate speech and general misinformation, and build their data by translating English datasets that contain no South African election content.

(ii) The only design that links automated flagging to human verification on a ledger [38] is conceptual and targets an Ethereum-compatible network with token incentives, not a consortium of known institutions.

(iii) No reviewed system binds roles to attributes in participants' X.509 certificates evaluated inside chaincode, and the read-only roles for external parties that some systems propose are not enforced by the ledger itself.

(iv) No reviewed system decides what to store on-chain with reference to POPIA.

(v) No misinformation-specific system reports measured latency or throughput. The measurements available in related domains [90], [95] come from systems without a detection stage.

This study addresses each of these gaps with the artefact described in Section 3, from the election-specific corpus and AfroXLMR detector (Sections 3.1 and 3.2) to the certificate-based Fabric consortium, its independent observer role and its end-to-end and Caliper benchmarks (Sections 3.3 to 3.7).

6 Discussion

6.1 AI-Enabled Misinformation Detection

This section discusses the results of the fine-tuning study described in Section 3.2. All figures are measured on the held-out test set of 1,393 statements, and the transformer results are summarised over the three seeds (619, 436 and 777). Table 12 compares all models, Table 13 gives the full results for AfroXLMR-large-76L, and Table 14 breaks its F1-score down by language.

Table 12: Test-set results by model and fine-tuning method. Transformer figures are means over three seeds, with the sample standard deviation of F1; the classical baselines are deterministic.

Model	Method	Accuracy	F1	AUC
TF-IDF + logistic regression	Classical	0.799	0.807	0.886
TF-IDF + linear SVM	Classical	0.814	0.822	0.899
TF-IDF + multinomial NB	Classical	0.816	0.827	0.898
MzansiLM-125M	Full	0.842	0.851 ± 0.003	0.918
MzansiLM-125M	LoRA	0.773	0.788 ± 0.003	0.863
AfroXLMR-base-76L	Full	0.851	0.859 ± 0.005	0.935
AfroXLMR-base-76L	LoRA	0.815	0.826 ± 0.010	0.901
AfroXLMR-large-76L	Full	0.895	0.899 ± 0.009	0.961
AfroXLMR-large-76L	LoRA	0.877	0.882 ± 0.015	0.940

6.1.1 Overall Performance

AfroXLMR-large-76L, the model retained for the framework (Section 3.2), achieved the highest F1-score under both fine-tuning methods: 0.899 ± 0.009 with full fine-tuning and 0.882 ± 0.015 with LoRA. Its best single run, full fine-tuning with seed 777, reached an F1-score of 0.906 and an AUC of 0.963. Every AfroXLMR configuration outperformed the three TF-IDF baselines (F1 0.807–0.827), of which multinomial naive Bayes was the strongest. This differs from the South African studies reviewed in Section 5.1, where classical classifiers performed best and the one transformer evaluated, trained on 1,500 translated sentences, reached at most 79% accuracy [15]; the larger election corpus used here (11,099 training statements) may explain why the transformers pull ahead. The ranking is consistent with earlier findings that fine-tuned encoders, and Afrocentric encoders in particular, outperform decoder-only models on classification [24], [26], [87]. MzansiLM-125M, the only model built specifically for South African languages, came close to AfroXLMR-base under

full fine-tuning (0.851 against 0.859), but under LoRA it was the only transformer configuration that failed to beat the TF-IDF baselines (0.788).

6.1.2 Full Fine-tuning Versus LoRA

Full fine-tuning achieved the higher F1-score in all nine comparisons (three models, three seeds), but the margin narrowed as the models grew: 0.063 for MzansiLM-125M, 0.033 for AfroXLMR-base and 0.018 for AfroXLMR-large. On the largest model LoRA therefore recovered about 98% of the fully fine-tuned F1-score while training 1.29% of the parameters (Table 8), whereas on MzansiLM-125M its cost was large. This supports measuring the difference between the methods rather than assuming they are equivalent [28]. LoRA was also less consistent across seeds: its F1 standard deviation was higher on both AfroXLMR models (0.010 against 0.005, and 0.015 against 0.009) and equal on MzansiLM-125M. On AfroXLMR-large the single-seed gap between the methods ranged from 0.002 (seed 619) to 0.038 (seed 436), and LoRA's own spread across seeds (0.865–0.893) exceeded the mean gap, so a comparison based on any one seed would have understated or overstated the difference. This confirms the value of training with several seeds [30].

Table 13: Test-set results for AfroXLMR-large-76L by fine-tuning method and seed.

Method	Seed	Accuracy	Precision	Recall	F1	AUC
Full	619	0.881	0.841	0.943	0.889	0.956
Full	436	0.901	0.890	0.918	0.903	0.963
Full	777	0.902	0.875	0.939	0.906	0.963
Full	Mean	0.895	0.869	0.933	0.899	0.961
LoRA	619	0.885	0.885	0.888	0.887	0.946
LoRA	436	0.858	0.833	0.899	0.865	0.929
LoRA	777	0.889	0.869	0.918	0.893	0.945
LoRA	Mean	0.877	0.862	0.902	0.882	0.940

6.1.3 Training Behaviour

The two methods also trained differently. Under full fine-tuning, the epoch selected by early stopping came early (on average epoch 3.0 to 5.0, depending on the model), and by then the validation loss was already well above the training loss, by up to 0.96 on MzansiLM-125M and by 0.36–0.58 on AfroXLMR-base, a sign of overfitting. LoRA runs peaked later (on average epoch 5.3 to 7.0) and stopped with training and validation losses close together (differences between −0.04 and +0.13). LoRA thus gives up a small amount of F1 for a much

smaller gap between training and validation loss, although its results vary more between seeds.

6.1.4 Performance by language

AfroXLMR-large-76L performed best on English and slightly less well, and almost equally, on isiZulu and Sepedi (Table 14). The gap between English and the two African languages, about 0.04 F1 under full fine-tuning, is modest, which satisfies the selection criterion that performance on Sepedi and isiZulu should not fall far behind English (Section 3.2). Unlike the sentiment results of Mabokela et al., in which AfroXLM-R performed better on the Nguni languages than on the Sotho-Tswana group [70], Sepedi and isiZulu scored almost identically here; one possible explanation is that the 76-language AfroXLMR used in this study includes both languages in pre-training. The cost of LoRA was slightly larger on the African languages (0.020) than on English (0.016).

Table 14: F1-score of AfroXLMR-large-76L by language, mean over three seeds.

Method	English	Sepedi	isiZulu	Sepedi and isiZulu
Full	0.926	0.885	0.889	0.887
LoRA	0.910	0.865	0.870	0.867
Difference (Full − LoRA)	0.016	0.020	0.019	0.020

6.1.5 Errors and triage

For a triage policy that escalates only predicted misinformation, a missed misinformation statement would not reach a fact-checker and a false positive would consume reviewer time. The feed benchmark instead reviewed both predicted labels, so its workflow completion results do not evaluate that escalation policy. In the fine-tuning study, full fine-tuning's advantage on AfroXLMR-large came mainly from recall (mean 0.933 against 0.902 for LoRA), with similar precision (0.869 against 0.862). Of the 704 misinformation statements in the test set, full fine-tuning missed between 40 and 58 per seed and LoRA between 58 and 79. On seed 619, where their F1 scores were nearly tied, full fine-tuning raised more false positives (126 against 81).

6.1.6 Choice of deployed model

The framework deploys AfroXLMR-large-76L fine-tuned with LoRA rather than the fully fine-tuned model. On the synthetic test set LoRA trails full fine-tuning by 0.018 F1, mainly through lower recall (0.902 against 0.933), and its results vary more between seeds. Its training behaviour, however, makes it the safer choice for live use. Two of the three fully

fine-tuned AfroXLMR-large runs stopped with the validation loss 0.40–0.49 above the training loss, a sign that the model had begun to fit the training data itself, whereas all three LoRA runs stopped with the two losses within 0.04 of each other. Because the test set is produced by the same synthetic pipeline as the training data, a model that has overfitted that data can still score well on it; on real, unseen election posts, which differ in vocabulary, errors and style (Section 3.1.1), the LoRA model is expected to generalise better. LoRA also suits deployment: the adapter trains 1.29% of the model's parameters and is loaded on a shared base model (Section 3.2).

6.1.7 Limitations

These results are measured on synthetic statements (Section 3.1.1), the scores may therefore not transfer directly to real election posts, and whether the deployed LoRA model generalises better than full fine-tuning on such posts, as its training behaviour suggests, has still to be established.

6.2 Blockchain Layer

6.2.1 Primary Benchmark: Throughput and Latency

The primary benchmark evaluated the complete feed workflow at consortium sizes of 3, 5, 10, 15 and 20 organisations, with 150 recorded workflow observations per size. All 750 observations reached FINAL and passed integrity verification, with no failed sample workflows. Table 15 separates claim-commitment latency, final-verdict latency and the subsequent integrity-check duration.

Consortium Size	Observations/Samples	Mean Ingestion to Committed Claim (ms)	Mean Integrity Verification (ms)	Mean Ingestion to Final Verdict (ms)
3	150	2,389.117	57.031	6,692.853
5	150	2,411.698	67.829	8,904.244
10	150	2,413.322	78.714	13,163.842
15	150	2,435.568	95.488	19,759.058
20	150	2,431.033	98.469	24,093.229

Table 15: Mean feed-workflow timings by consortium size under sequential scripted review. Durations are milliseconds, the count is workflow observations.

Note: Ingestion-to-final-verdict includes the scripted review sequence and excludes the integrity-verification request.

6.2.1.1 Claim commitment and payload scope

Mean ingestion to committed claim latency varied from 2,389.117 to 2,435.568 ms across the five consortium sizes. This relatively narrow range concerns the initial claim commitment, not completion of the whole review workflow. Keeping claim text off chain limits the claim payload, but the experiment did not isolate payload size as the cause of the observed timings.

The report's stored reviewer set, and vote records also depend on the workflow, so fixed claim content does not imply a constant cost for all review activity.

6.2.1.2 Endorsement and ordering configuration

The runs held the two-organization endorsement policy and single-orderer configuration constant while changing the number of provisioned organizations. Increasing consortium membership therefore did not increase the required endorsement count in these tests. This helps distinguish the initial commitment stage from the application-level review threshold. The measurements do not separately identify the contributions of peer selection, ordering, block formation or host contention, and do not test a distributed ordering service.

6.2.1.3 Sequential review overhead

Mean ingestion to final verdict latency increased from 6,692.853 ms at 3 organizations to 24,093.229 ms at 20, an increase of 17,400.376 ms, or approximately 3.6 times the initial duration. Since votes were submitted sequentially until a majority formed, a larger consortium required more sequential review transactions before finalization. This pattern is consistent with additional review overhead, although the experiment does not isolate the latency of one vote from the other workflow costs. Mean integrity verification duration also increased, from 57.031 to 98.469 ms. The evidence supports relatively stable initial claim commitment alongside increasing final verdict latency, it does not support flat end-to-end scaling or a marginal vote cost of 1–2 ms.

6.2.2 Secondary Benchmark: Throughput and Latency

Hyperledger Caliper directly exercised a counted write round followed by a counted full-scan read round against each provisioned network. The benchmark covered 3, 5, 10, 15 and 20 organizations and per-round counts of 100, 250, 500, 1,000 and 2,000 transactions. Two workers shared each round's total count, with a fixed-load target of 100 pending transactions. Send rates were measured rather than fixed at 25 or 50 TPS. All 25 selected configurations passed, and all 25 write rounds and 25 read rounds reported zero failures.

Tables 16 and 17 show arithmetic mean across the five consortium sizes at each transaction count. These are means across different membership configurations. The maximum and minimum latency columns are the five round maxima and minimum. Average latency is the arithmetic mean of Caliper's five reported round average latencies. The detailed per-consortium results are retained in the supplementary benchmark table.

Transactions per round	Mean success	Mean fail	Mean send rate (TPS)	Mean of round max latency (s)	Mean of round min latency (s)	Mean round-average latency (s)	Mean throughput (TPS)
100	100	0	142.26	0.538	0.098	0.320	86.92
250	250	0	133.80	0.758	0.096	0.474	114.86
500	500	0	140.56	0.834	0.090	0.458	130.82
1,000	1,000	0	142.96	0.954	0.098	0.496	139.16
2,000	2,000	0	138.88	1.444	0.098	0.512	136.52

Table 16: Caliper write-round results, averaged across five consortium sizes. Counts are attempted transactions per round, shared across two workers, latencies are second and rates are TPS.

All measured write rounds completed without reported failures. Mean throughput across consortium sizes increased from 86.92 TPS at 100 transactions to 139.16 TPS at 1,000 and was 136.52 TPS at 2,000. Mean round-average latency ranged from 0.320 to 0.512 seconds. The detailed 2,000 transaction rounds show that write throughput was 185.1 TPS at 3 organizations

and 74.8 TPS at 20, with average latency increasing from 0.36 to 0.89 seconds. Therefore, the results do not support a claim that consortium size had no material association with write performance.

Transactions per round	Mean success	Mean fail	Mean send rate (TPS)	Mean of round max latency (s)	Mean of round min latency (s)	Mean round-average latency (s)	Mean throughput (TPS)
100	100	0	371.52	0.160	0.016	0.094	328.60
250	250	0	232.90	0.510	0.032	0.254	218.24
500	500	0	127.76	1.052	0.052	0.546	118.98
1,000	1,000	0	58.98	2.180	0.090	1.160	57.36
2,000	2,000	0	27.32	6.018	0.200	2.536	27.02

Table 17: Caliper full-scan read-round results, averaged across five consortium sizes. Counts are attempted transactions per round, shared across two workers; latencies are second and rates are TPS.

Read throughput declined and latency increased as the counted full-scan workload grew. Mean throughput across consortium sizes fell from 328.60 TPS at 100 requests to 27.02 TPS at 2,000, while mean round-average latency increased from 0.094 to 2.536 seconds. Every request in the selected rounds succeeded, including those at 1,000 and 2,000 requests. Consequently, this campaign shows degradation in completion rate and latency under the measured workload, rather than the previously reported 81.2% failure rate or an absolute collapse at 1,000 requests under CouchDB.

6.3 POPIA Compliance

Four key System design choices made to that align with the POPIA (Protection of Personal

Information Act) Act. These are discussed below:

6.3.1 POPIA Act, Section 10: Minimality

The First design choice is the principle of data minimization where only a SHA-256 content hash, CID (Content Identifier), AI verdict metadata, and organizational MSP identifier are written on-chain with the full claim text and evidence live off-chain in IPFS (Interplanetary File System).

6.3.2 POPIA Act, Section 13: Collection for specific purpose

The second design choice was purpose limitation which involves the on-chain schema being fixed by the ledger's own contract definitions, which allows for no free form or re-purposable personal fields existence on-chain

6.3.3 POPIA Act, Section 14: Retention and restriction of records

The third design choice was storage limitation whereby the Off-chain IPFS storage is implemented for the retention of off-chain documents governed by the IPFS node operator's own policy rather than by the chaincode.

6.3.4 POPIA Act, Section 19: Security measures on integrity and confidentiality of personal information

The final design choice made for alignment was the following of the principle of integrity and confidentiality. The principles were implemented in the form of the implementation of SHA-256 content hashing alongside content addressing giving a two layer tamper-resistance check.

Also, the network's TLS-secured, certificate authenticated channel provides transport security and role based access control (RBAC), enforced directly in the ledger's contract logic via certificate role attributes, restricting which identities may submit, fact check, finalize or govern the consortium.

7 Conclusion & Future Work

7.1 Conclusion

This study set out to determine whether a blockchain-based system combined with a multilingual transformer based AI classifier could provide a transparent, tamper-evident and role restricted framework for monitoring election related misinformation across three South African languages (English, Sepedi and isiZulu). By designing, implementing, and empirically evaluating this framework, the study successfully completed its defined core milestones (Objectives 3, 4, and 6).

7.1.1 Multilingual detection:

No labelled corpus of South African election misinformation existed in Sepedi or isiZulu (Objective 1), so the study generated an election-specific English corpus with a large language model and translated it into Sepedi and isiZulu, yielding 13,997 statements balanced across the three languages and the two labels. On this corpus, three TF-IDF baselines and three transformers were compared, with each transformer fine-tuned both fully and with LoRA and trained with three seeds (Objective 2). AfroXLMR-large-76L achieved the highest mean test F1-score under both methods, 0.899 with full fine-tuning and 0.882 with LoRA, against 0.807–0.827 for the baselines (Objective 7). Its performance held across languages: under full fine-tuning it scored 0.926 on English, 0.885 on Sepedi and 0.889 on isiZulu (0.910, 0.865 and 0.870 under LoRA), so the two African languages trailed English by about
0.04 and scored almost identically to each other.

The framework deploys the LoRA variant of AfroXLMR-large-76L. It trails full fine-tuning by 0.018 F1, mainly through lower recall, but all three LoRA runs stopped with training and validation losses within 0.04 of each other, whereas two of the three fully fine-tuned runs showed signs of overfitting, and the adapter trains only 1.29% of the model's parameters on a shared base model. The model acts as a triage tool rather than a judge: its label and confidence are recorded on the ledger alongside the fact-checkers' votes, not in place of them. These results answer SRQ1 for the conditions tested, showing that a fine-tuned Afrocentric encoder trained on LLM-generated, machine-translated data can detect election misinformation in Sepedi and isiZulu at close to its English performance. Because the training and test data are both synthetic, how well the model performs on real election posts has still to be established.

7.1.2 Transparent, immutable reporting:

To balance data integrity, privacy, accessibility, and scalability, a dedicated permissioned architecture was designed (Objective 3). Reports are anchored on-chain by SHA-256 content hash and structured metadata, with the full claim text held off-chain in a three node IPFS swarm and addressed by content identifier. This hybrid, privacy preserving on-chain/off-chain data model was specifically implemented to achieve compliance with South Africa's Protection of Personal Information Act (Objective 5). This data model satisfies POPIA's minimality and collection for specific purpose principles at the architectural level, while the on-chain SHA-256 hash provides tamper-evidence without exposing sensitive claim content to all network participants. A verify endpoint confirms integrity by comparing the off-chain document's SHA-256 hash against the on-chain record, validating the end-to-end, tamper-resistant monitoring workflow (Objective 4).

7.1.3 Role restricted governance:

Enforcing strict governance considerations (Objective 8), Role-Based Access Control (RBAC) was built directly into the ledger's contract logic through X.509 certificate role attributes. The system allows for three primary roles, those bing official, fact checker and observer. These roles ensure that only appropriately authorized identities may submit claims, cast fact-checks, finalize or expire reports and govern consortium membership. This check is performed inside the chaincode itself using the client identity library, providing a materially stronger access control guarantee than gateway level enforcement alone.

7.1.4 Scalability and performance:

The feed benchmark completed and verified 150 workflow observations at each of five consortium sizes, for 750 observations overall. Mean ingestion-to-committed-claim latency remained approximately 2.39–2.44 seconds. In contrast, mean ingestion-to-final-verdict latency increased from 6.69 seconds at 3 organisations to 8.90, 13.16, 19.76 and 24.09 seconds at 5, 10, 15 and 20 organisations, respectively. The required agreeing-vote count increased from two to ten, and votes were submitted sequentially until the majority threshold was reached. The evidence therefore supports relatively stable initial commitment with increasing final-verdict latency, rather than flat end to end scaling. Integrity verification, measured separately, increased from approximately 57 to 98 ms. These observations use the combined-run accounting described in Section 3.7.

The direct Caliper benchmarks covered five organization counts and five transaction counts from 100 to 2,000 per round, using two workers and a target of 100 pending transactions. Every selected write and read round completed without reported failures. Average across consortium sizes write throughput was 86.92 TPS at 100 transactions, 139.16 TPS at 1,000 and 136.52 TPS at 2,000. Mean round-average write latency ranged from 0.320 to 0.512 seconds. The per-consortium results show differences with membership size, for example at 2,000 transactions write throughput was 185.1 TPS for 3 organizations and 74.8 TPS for 20.

Full scan reads showed increasing latency and decreasing throughput as the measured workload grew, while retaining zero reported failures through 2,000 requests. Across consortium sizes, mean read throughput decreased from 328.60 to 27.02 TPS and mean round-average latency increased from 0.094 to 2.536 seconds between 100 and 2,000 requests. Initial and post write ledger cardinalities varied, so these are workload observations rather than a controlled comparison at identical database size. The campaign does not establish an absolute collapse index or maximum system capacity. The single-host setting, scripted reviews, synthetic inputs and finite counted rounds limit generalization to a distributed production deployment.

7.1.5 Overall

This study designed and evaluated a single-host prototype that combines multilingual claim classification, off-chain evidence storage and a permissioned Hyperledger Fabric ledger. The implemented workflow records claim and review evidence, enforces reviewer eligibility and role restrictions, forms automatic majority outcomes and verifies content commitments. All recorded feed-workflow observations and selected direct Caliper rounds completed successfully under the stated configurations. The updated results distinguish relatively stable claim commitment from increasing final-verdict latency as the sequential review threshold grows and show declining full-scan read throughput at larger measured workloads without the previously claimed collapse. These findings demonstrate prototype workflow feasibility under synthetic, scripted conditions.

7.2 Future Work

Within the confines of the project scope a few elements remained to be out of scope and hence proposed as future work/direction for the system:

7.2.1 Multi-Host and Inter-Institution Deployment

Extend the benchmark suite to a physically distributed, multi-host deployment with isolated resources for each organization. Measure claim commitment, sequential and concurrent review, integrity verification, and direct ledger writes and reads separately. Hold endorsement settings, initial ledger cardinality and workload controls constant when comparing consortium sizes and repeat configurations to quantify variability.

7.2.2 Stakeholder Web Interface for Workflow Execution

Build a stakeholder facing web interface on top of the existing REST gateway. A general-purpose blockchain explorer currently provides only a read only operational view, so the register, submit, vote, finalize and query workflow can only be exercised by a non-technical fact checker once such an interface exists.

7.2.3 Expanding Identity Management for Production Deployment

At present the optional JWT flow identifies only the organisation, not the individual fact checker. Future work should embed per-user claim (user ID, role, organisation MSP ID, and permitted actions) and bind them to a Fabric identity or certificate attribute evaluated by the chaincode. Authorisation should remain in chaincode, while the gateway continues to authenticate and route requests. This would enable individual accountability for submissions, votes, and finalisations. It should also add token expiry, refresh token support, revocation and key rotation to both JWT key mode. This remains future work because the present study focuses on organisation level behaviour and treats JWT as future expansion for a future production ready system deployment

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